Power Amplifiers

Classification Of Amplifiers

1. According to frequency capabilities.

Amplifiers are classified as audio amplifiers , radio frequency amplifiers

- **AF Amplifier** are used to amplify the signals lying in the audio range (i.e. 20 Hz to 20 kHz)
- **RF amplifiers** are used to amplify signals having very high frequency.

2. According to coupling methods.

- R-C coupled amplifiers,
- Transformer coupled amplifiers
- Direct Coupled

Classification Of Amplifiers

3. According to use.

a. Voltage amplifiers

- Amplify the input voltage, if possible with minimal current at the output.
- The power gain of the voltage amplifier is low.
- The main application is to strengthen the signal to make it less affected by noise and attenuation.
- Ideal voltage amp. have infinite input impedance & zero output impedance.

a. Power amplifiers

- Amplify the input power, if possible with minimal change in the output voltage
- Power amp. are used in devices which require a large power across the loads.
- In multi stage amplifiers, power amplification is made in the final stages
  - ✓ Audio amplifiers and RF amplifiers use it to deliver sufficient power the load.
  - ✓ Servo motor controllers use power it to drive the motors.

Classification Of Amplifiers

	Voltage amplifiers	Power amplifiers
current gain	low	high
Voltage gain	high	low
Heat dissipation	low	high
cooling mechanism	not required	required
Transistor Size	Small	Large to dissipate heat
Base Width	small	Wide to handle higher current
Beta	Usually high >100	Low usually < 20

Basics ….

- In mathematical terms, if the input signal is denoted as **S**, the output of a *perfect* amplifier is **X*S**, where X is a *constant* (a fixed number). The "*" symbol means ”multiplied by".

- No amplifier does exactly the ideal .

Contd..

- But many do a very good job if they are operated within their advertised power ratings .

- Output signal of *all* amplifiers contain additional (unwanted) components that are not present in the input signal; these additional characteristics may be lumped together and are generally known as ***distortion*****.**

Contd….

- Power amplifiers get the necessary energy for amplification of input signals from the AC wall outlet to which they are plugged into.

- If you had a *perfect* amplifier, all of the energy the amplifier took from the AC outlet would be converted to useful output (to the speakers)

Contd ….

- Power is not really something that can be "amplified". *Voltage* and *current* can be amplified.

- The term "power amplifier" although technically incorrect has become understood to mean an amplifier that is intended to drive a load (such as a speaker, a motor, etc).

Functional blocks of an amplifier

- All power amplifiers have:

1. *A Power supply*

2. *An input stage*

3. *An output stage*

1.Power Supply

- The primary purpose of a power supply in a power amplifier is to take the 120 V AC power from the outlet and convert it to a DC voltage.

- The very best of amplifiers have two totally independent power supplies, one for each channel (they do share a common AC power cord though).

2. Input Stage

- The general purpose of the input stage of a power amplifier (sometimes called the "front end") is to receive and prepare the input signals for "amplification" by the output stage.

- Two types:

  1. Balanced Input
  2. Single Ended Input

2. Input Stage

- **Balanced inputs** are much preferred over **single ended inputs** when interconnection cables are long and/or subject to noisy electrical environments because they provide very good *noise rejection*.

- The input stage also contains things like input level controls.

3.Output Stage

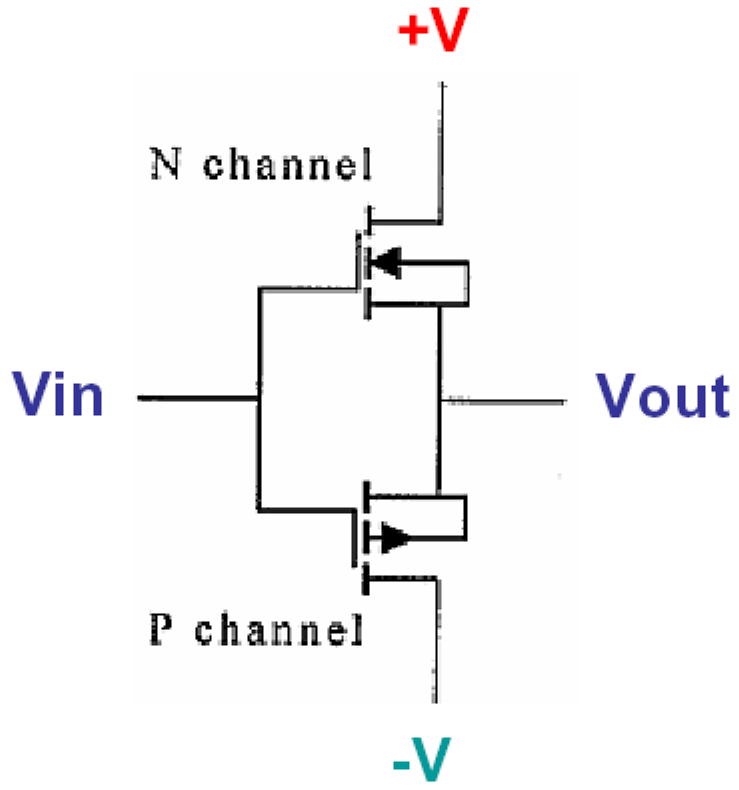
- The portion which actually converts the weak input signal into a much more powerful "replica" which is capable of driving high power to a speaker.
- This portion of the amplifier typically uses a number of "power transistors" (or MOSFETs) and is also responsible for generating the most **heat** in the unit.
- The output stage of an amplifier interfaces to the speakers.

Amplifier Classes

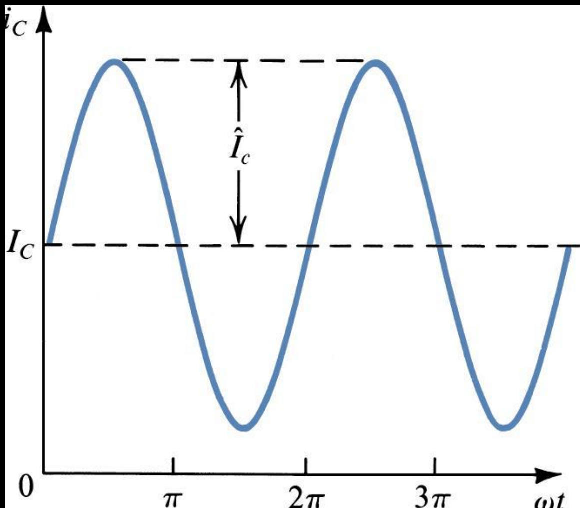
- The **Class** of an amplifier refers to the design of the circuitry within the amp.

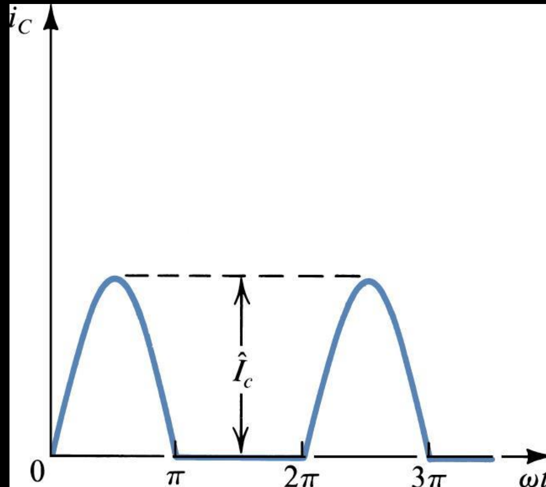
- For audio amplifiers, the **Class of amp refers to the output stage of the amp.**

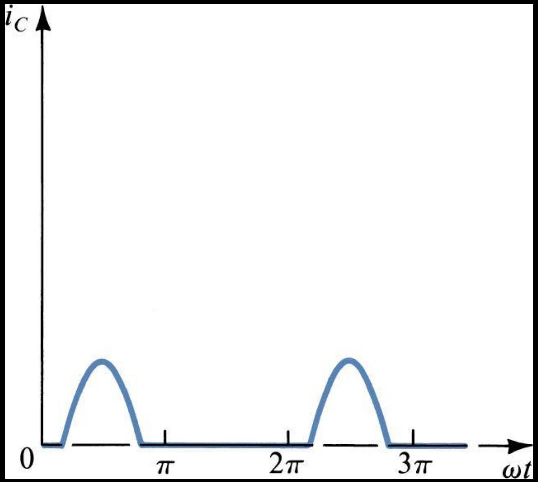
Basic Power Amplifier Output Stage
(Source-Follower MOSFET Configuration)

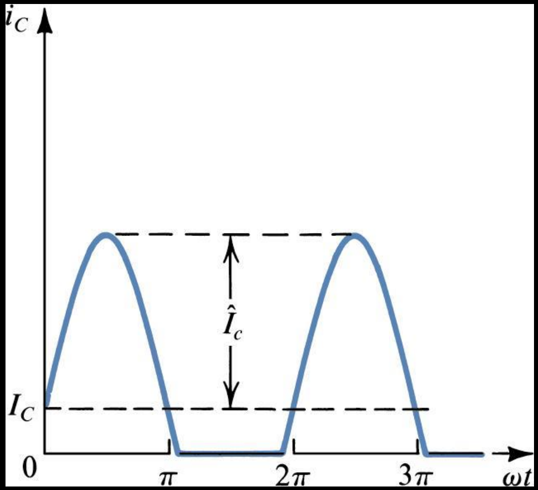


Classes









Collector current waveforms for transistors operating in (a) class A, (b) class B, (c) class AB, and (d) class C amplifier stages.

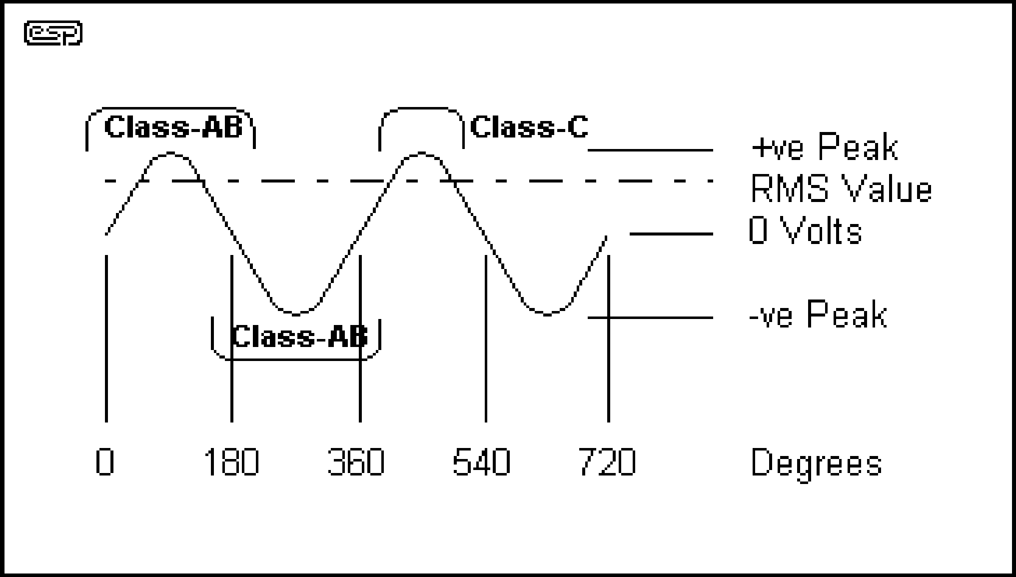
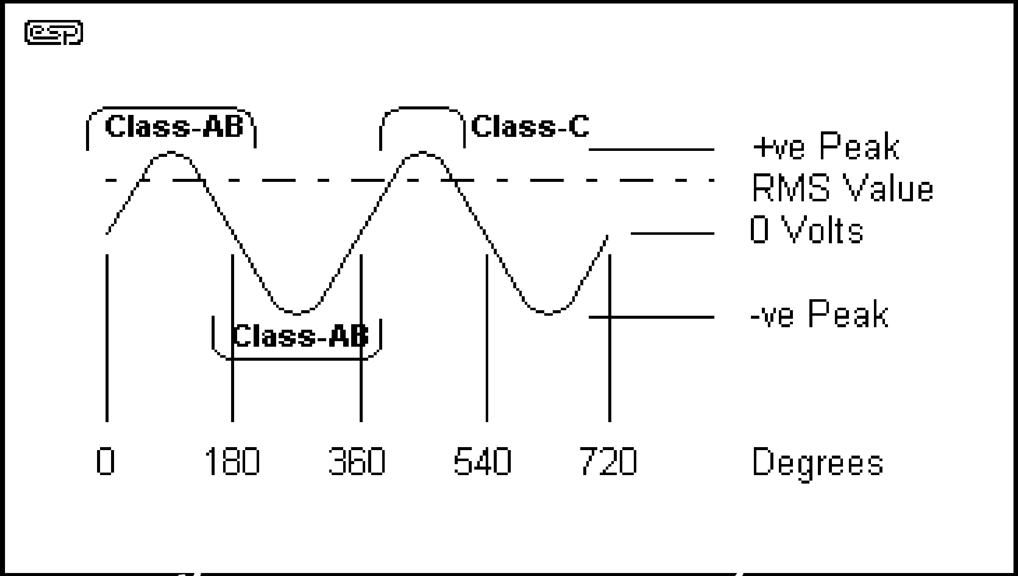


Figure 1 - The Sinewave Cycle

- **Class-A:** Output device(s) conduct through 360 degrees of input cycle (never switch off) - A single output device is possible. The device conducts for the entire waveform in Figure 1
- **Class-B:** Output devices conduct for 180 degrees (1/2 of input cycle) - for audio, two output devices in "push-pull" must be used (see Class-AB)
- **Class-AB:** Halfway (or partway) between the above two examples (181 to 200 degrees typical) - also requires push-pull operation for audio. The conduction for each output device is shown in Figure 1.



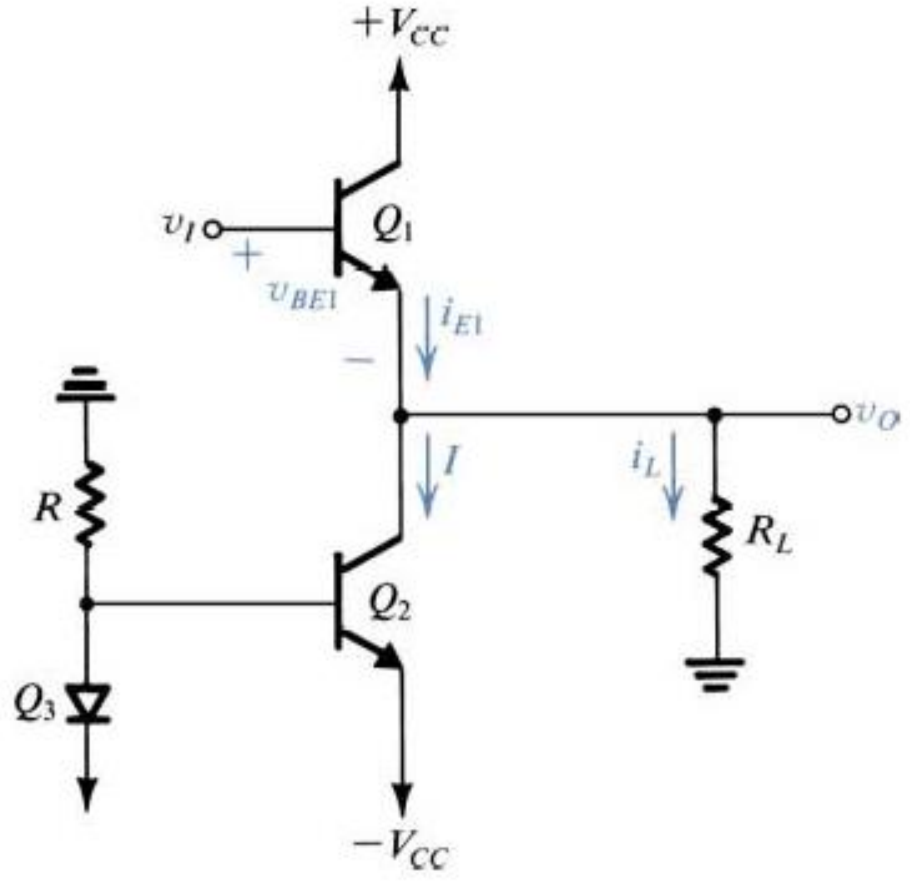
- **Class-C:** Output device(s) conduct for less than 180 degrees (100 to 150 degrees typical) - Radio Frequencies only - cannot be used for audio! This is the sound heard when one of the output devices goes open circuit in an audio amp! See Figure 1, showing the time the output device conducts

Power Amplifier Classes – "A"

- Class "A"
  - key ingredient of class A operation is that output device is always on
  - single-ended design with only one type polarity output device
  - the most inefficient of all power amplifier designs, averaging only around 20% (large, heavy, and run very hot)
  - are inherently the most linear, with the least amount of distortion

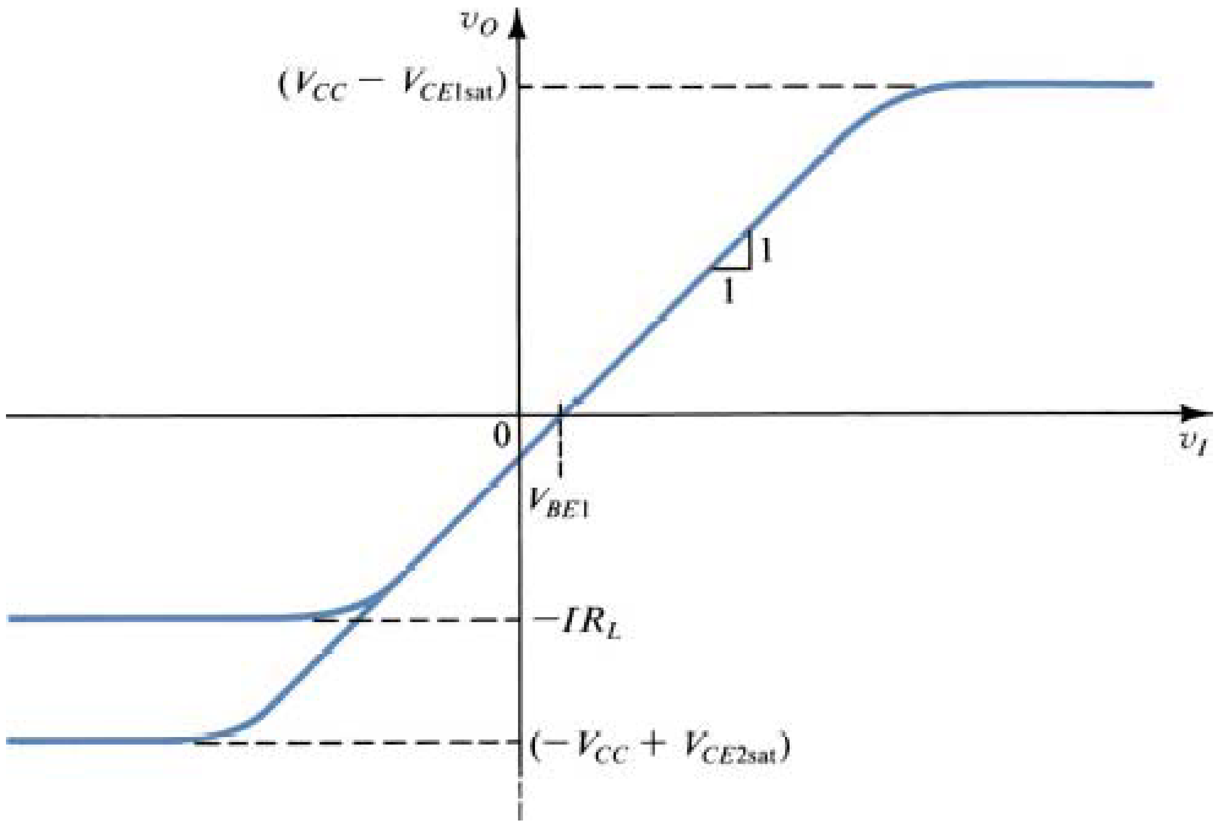
Class A Output Stage

- Class A output stage is a simple linear current amplifier.
- It is also very inefficient, typical maximum efficiency between 10 and 20 %.
- Only suitable for low power applications.
- High power requires much better efficiency.

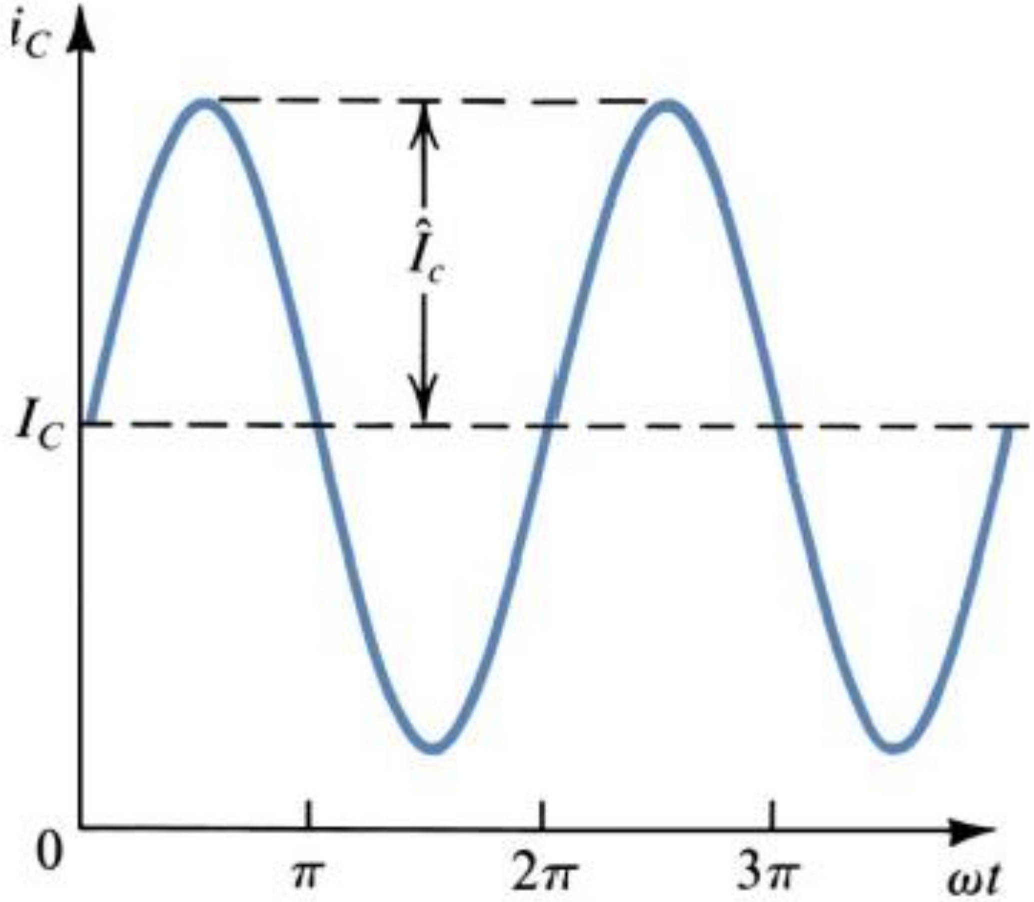


An emitter follower (Q_1) biased with a constant current I supplied by transistor Q_2.

Transfer Characteristics

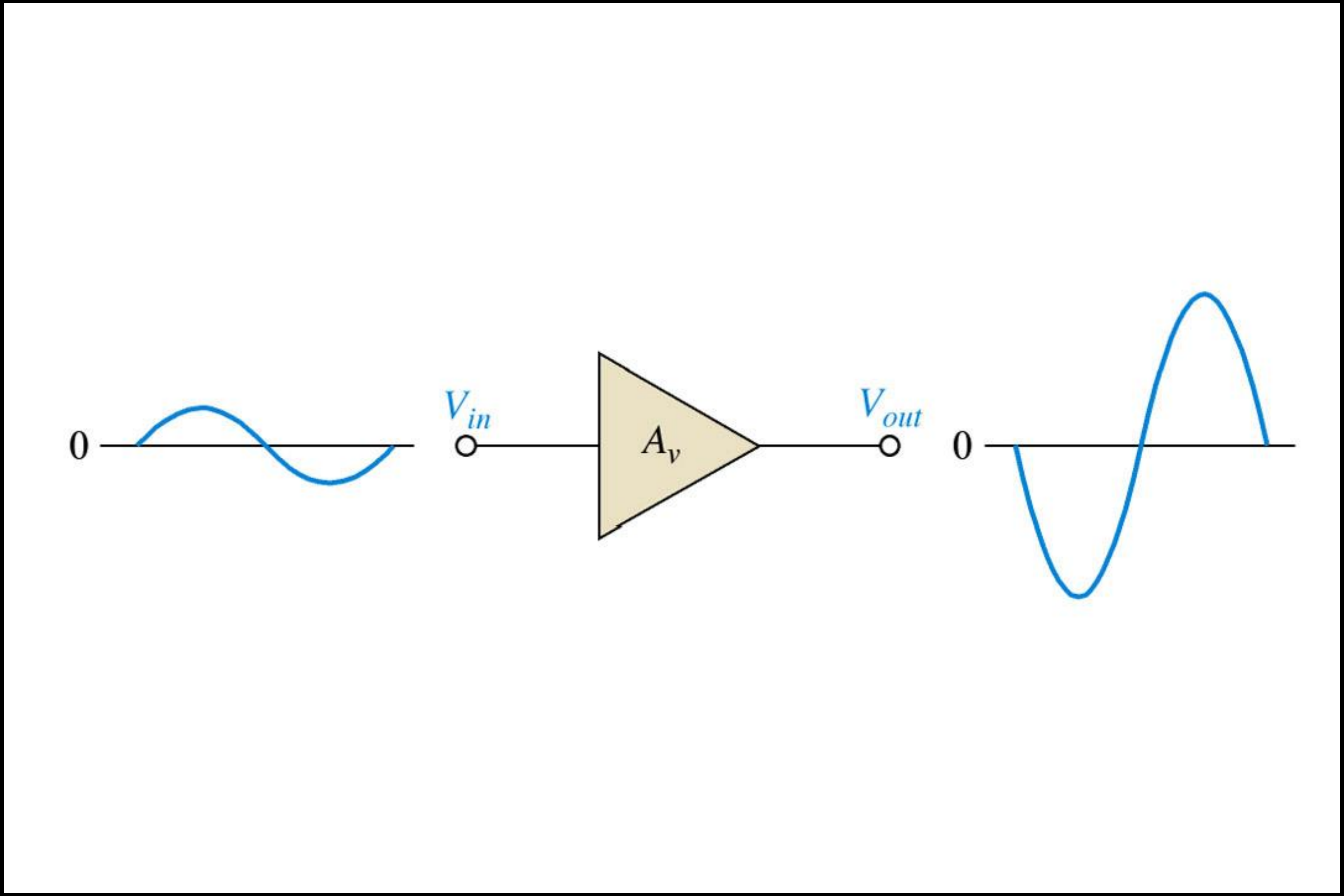


Transfer characteristic of the emitter follower. This linear characteristic is obtained by neglecting the change in v_{BE1} with i_L. The maximum positive output is determined by the saturation of Q_1. In the negative direction, the limit of the linear region is determined either by Q_1 turning off or by Q_2 saturating, depending on the values of I and R_L.

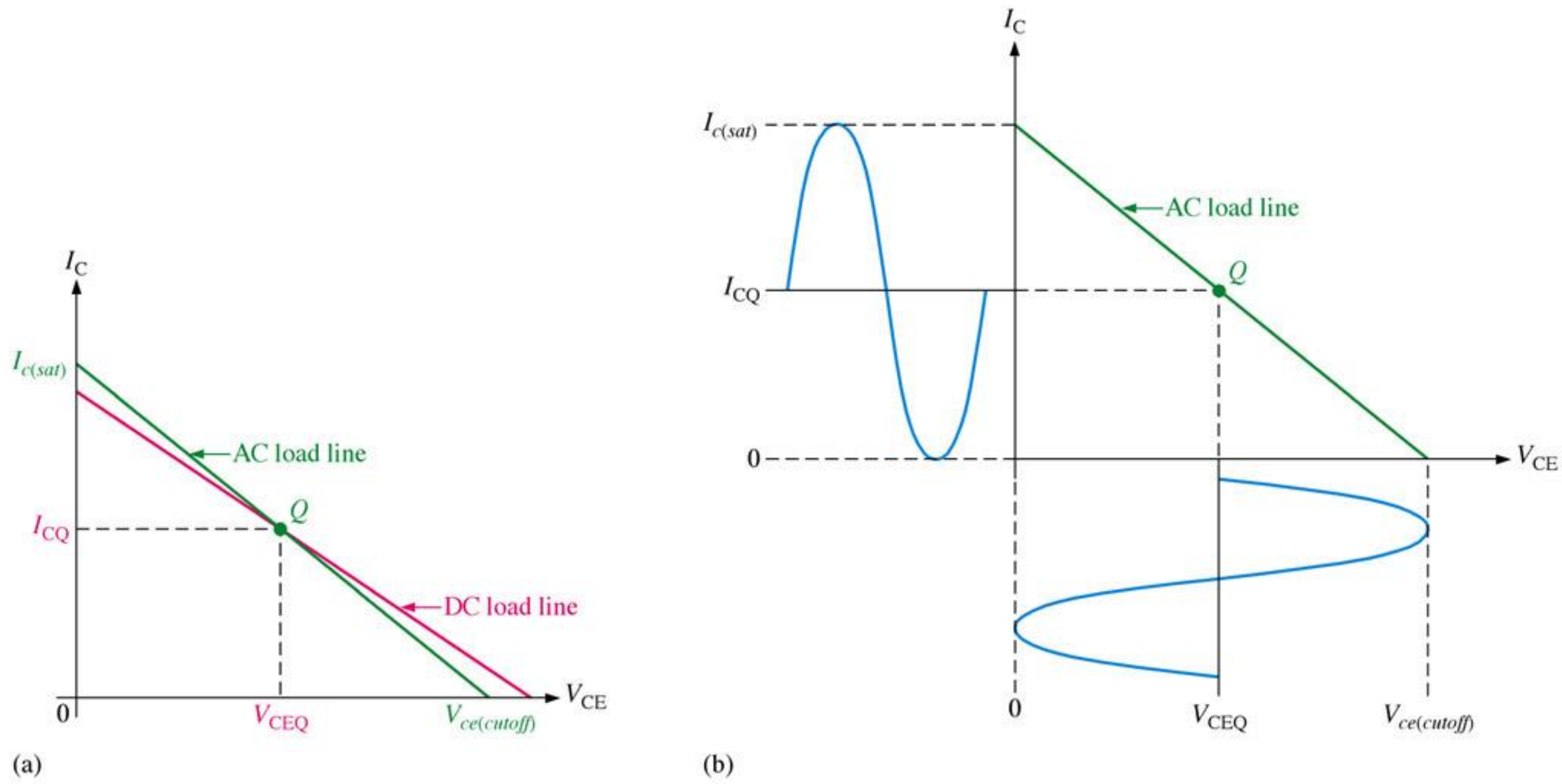


Collector current waveforms for transistors operating in class A amplifier stage

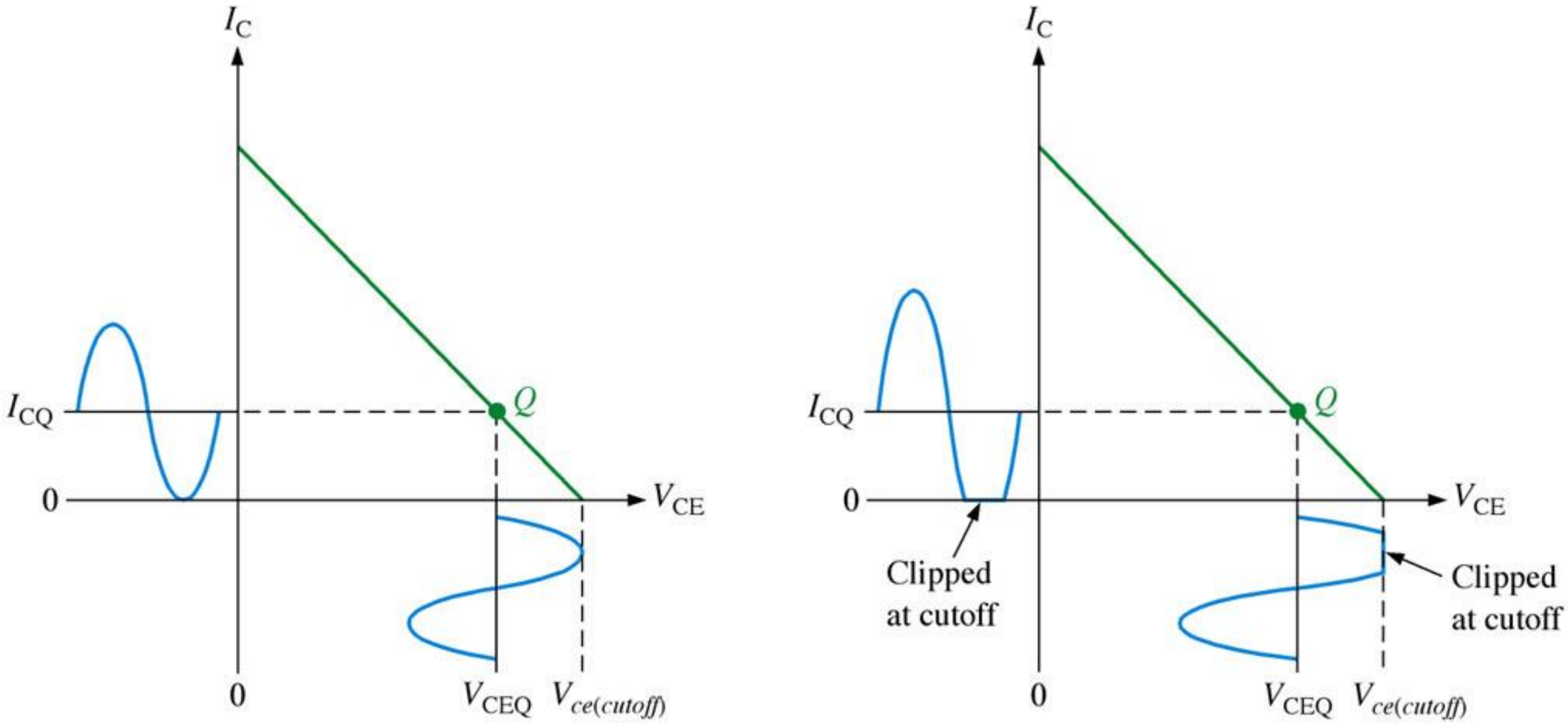
Basic class A amplifier operation. Output is shown 180∞ out of phase with the input (inverted).



Maximum class A output occurs when the Q-point is centered on the ac load line.



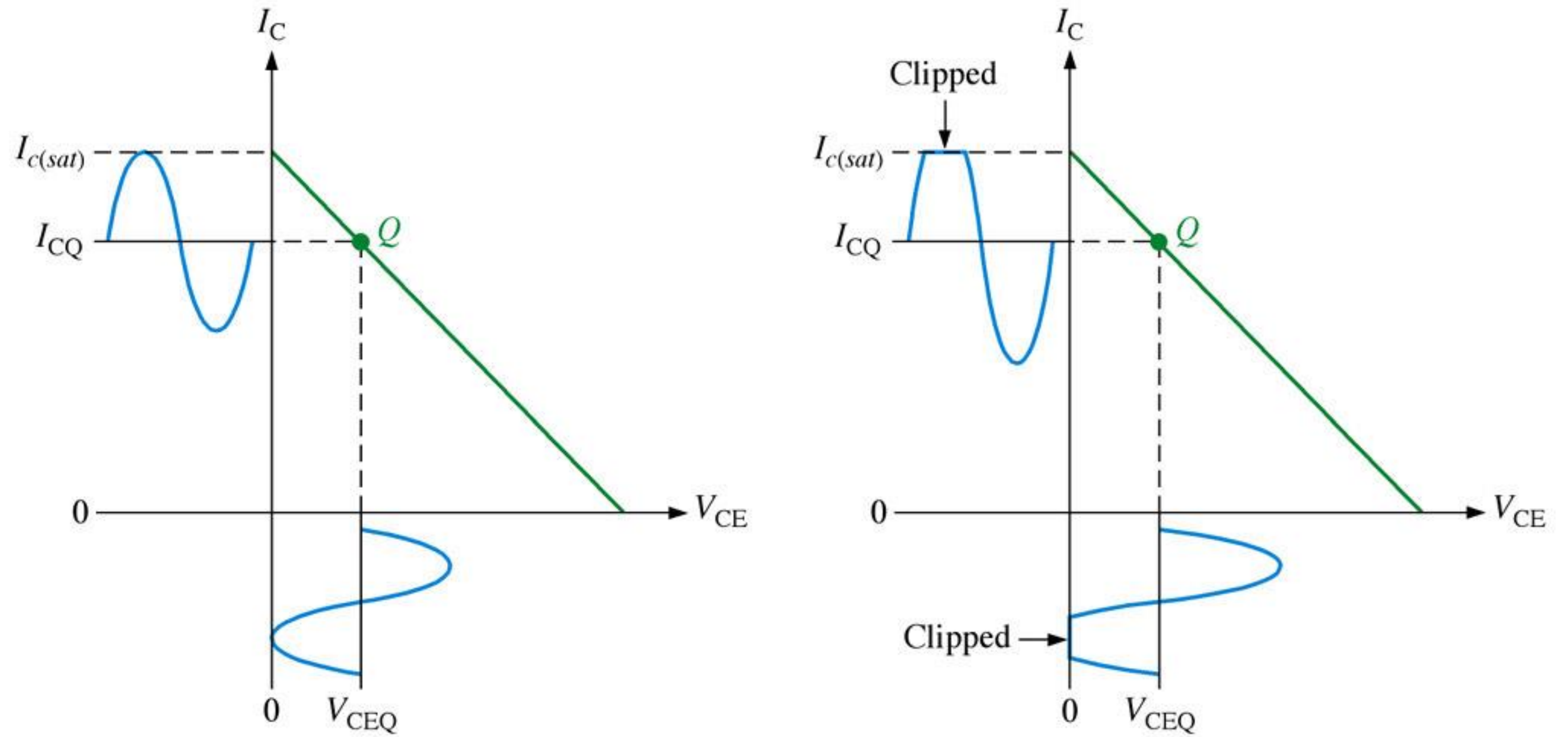
Q-point closer to cutoff.



(a) Amplitude of V_{ce} and I_c limited by cutoff

(b) Transistor driven into cutoff by a further increase in input amplitude

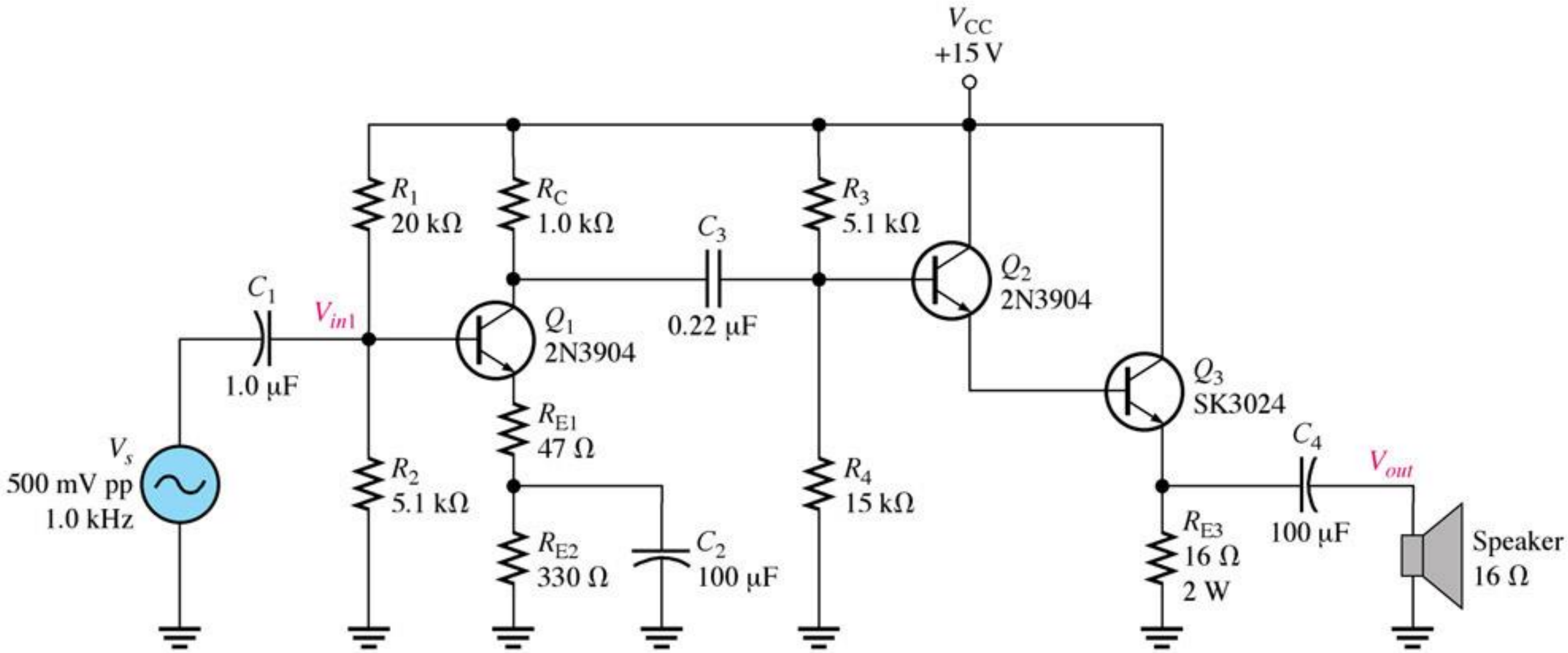
Q-point closer to saturation.



(a) Amplitude of V_{ce} and I_c limited by saturation

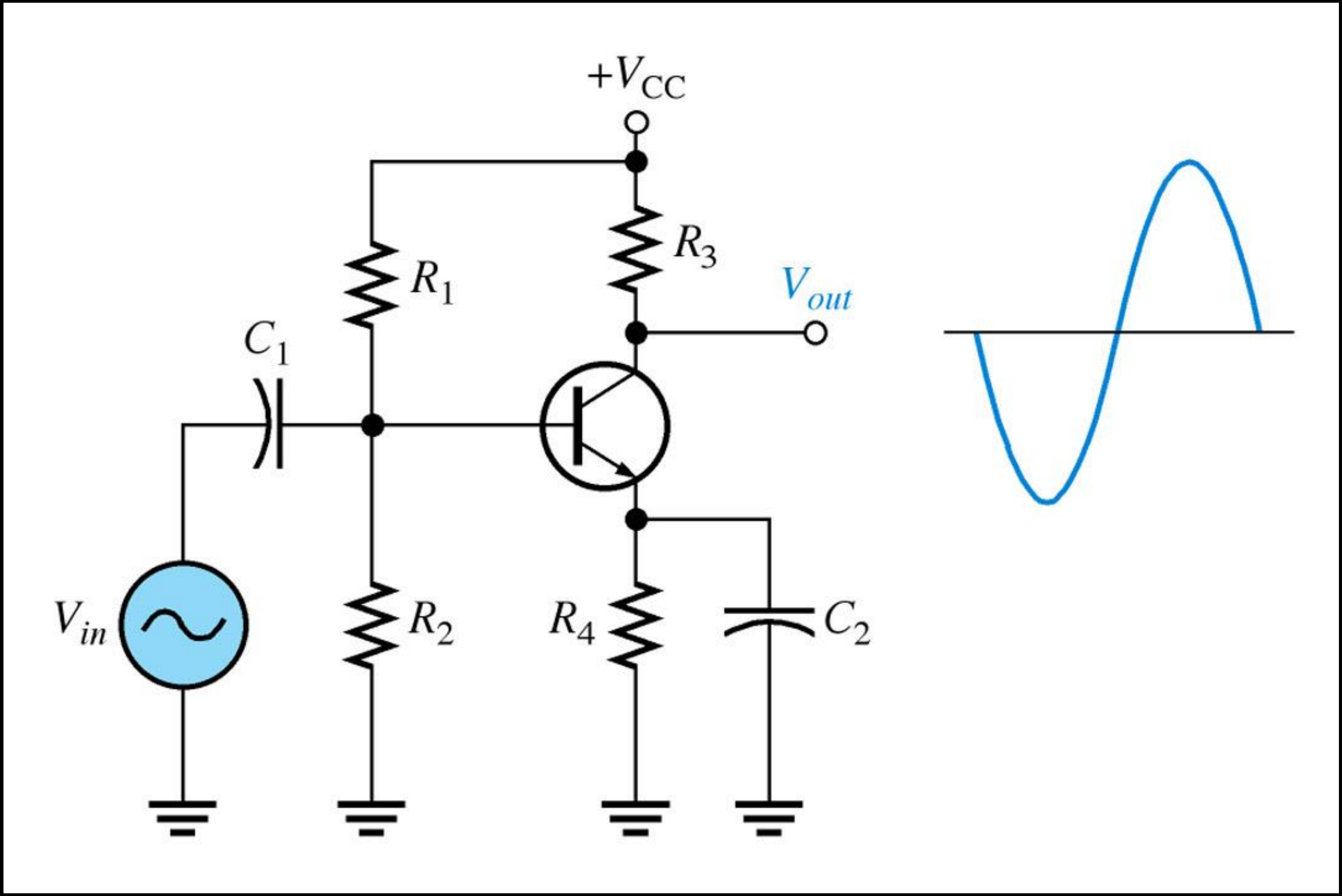
(b) Transistor driven into saturation by a further increase in input amplitude

FIGURE



V_{CC}
+15 V
R_1
20 kΩ
R_C
1.0 kΩ
C_3
0.22 µF
R_3
5.1 kΩ
Q_2
2N3904
C_1
1.0 µF
V_{in1}
Q_1
2N3904
Q_3
SK3024
V_s
500 mV pp
1.0 kHz
R_{E1}
47 Ω
R_2
5.1 kΩ
R_{E2}
330 Ω
C_2
100 µF
R_4
15 kΩ
C_4
100 µF
V_{out}
R_{E3}
16 Ω
2 W
Speaker
16 Ω

FIGURE 9-30 Class A power amplifier with correct output voltage swing.



SERIES-FED CLASS A AMPLIFIER

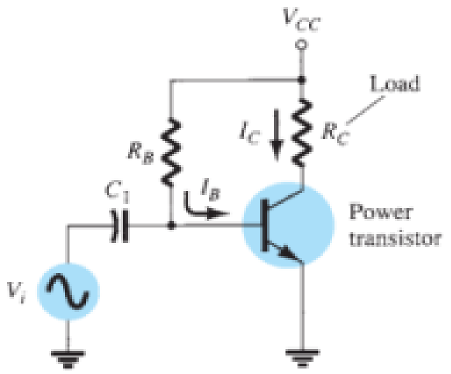


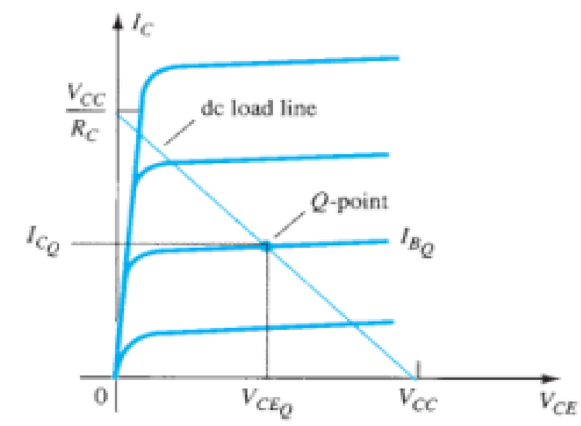
FIG. 12.2
Series-fed class A large-signal amplifier.

- DC Bias Operation

$$I_B = \frac{V_{CC} - 0.7\ \text{V}}{R_B}$$

$$I_C = \beta I_B$$

$$V_{CE} = V_{CC} - I_C R_C$$



- AC Operation

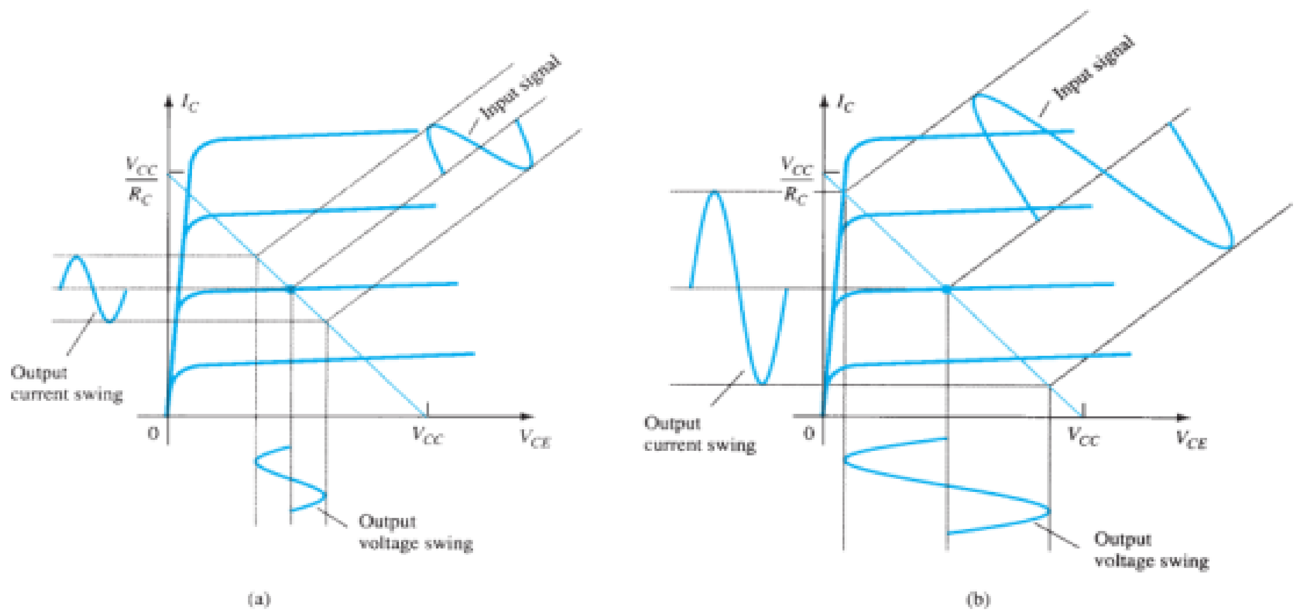


FIG. 12.4
Amplifier input and output signal variation.

Power Considerations

- The power drawn from the supply is $\boxed{P_i(\text{dc}) = V_{CC} I_{C_Q}}$

- Output Power

$$\boxed{P_o(\text{ac}) = V_{CE}(\text{rms}) I_C(\text{rms})}$$

$$\boxed{P_o(\text{ac}) = I_C^2(\text{rms}) R_C}$$

$$\boxed{P_o(\text{ac}) = \frac{V_C^2(\text{rms})}{R_C}}$$

- Efficiency

$$\boxed{\% \eta = \frac{P_o(\text{ac})}{P_i(\text{dc})} \times 100\%}$$

- Maximum Efficiency

$$\text{maximum } V_{CE}(\text{p-p}) = V_{CC}$$

$$\text{maximum } I_C(\text{p-p}) = \frac{V_{CC}}{R_C}$$

$$\text{maximum } P_o(\text{ac}) = \frac{V_{CC}(V_{CC}/R_C)}{8} = \frac{V_{CC}^2}{8R_C}$$

$$\text{maximum } P_i(\text{dc}) = V_{CC}(\text{maximum } I_C) = V_{CC}\frac{V_{CC}/R_C}{2} = \frac{V_{CC}^2}{2R_C}$$

The maximum power input can be calculated using the dc bias current set to one-half the

$$\text{maximum } \% \eta = \frac{\text{maximum } P_o(\text{ac})}{\text{maximum } P_i(\text{dc})} \times 100\% = \frac{V_{CC}^2/8R_C}{V_{CC}^2/2R_C} \times 100\% = 25\%$$

N.B.:

$$V_{\text{RMS}} = \frac{V_{\text{P}}}{\sqrt{2}}$$

Why is class A so inefficient ?

- Single transistor can only conduct in one direction.
- D.C. bias current is needed to cope with negative going signals.
- 75 % (or more) of the supplied power is dissipated by d.c.
- Solution : eliminate the bias current.

Class A

- Class A amplifiers have very low distortion (lowest distortion occurs when the volume is low)
- They are very inefficient and are rarely used for high power designs.
- The distortion is low because the transistors in the amp are biased such that they are half "on" when the amp is idling

Class A

- As a result of being half on at idle, a lot of power is dissipated in the devices even when the amp has no music playing!
- Class A amps are often used for "signal" level circuits (where power requirements are small) because they maintain low distortion.

Class-A Benefits

- The first is circuit simplicity.
- The signal is subjected to comparatively little amplification, resulting in an open loop gain which is generally fairly low.
- This means that very little overall feedback is used, so stability and phase should be excellent over the audio frequencies.
- Do not require any frequency compensation.

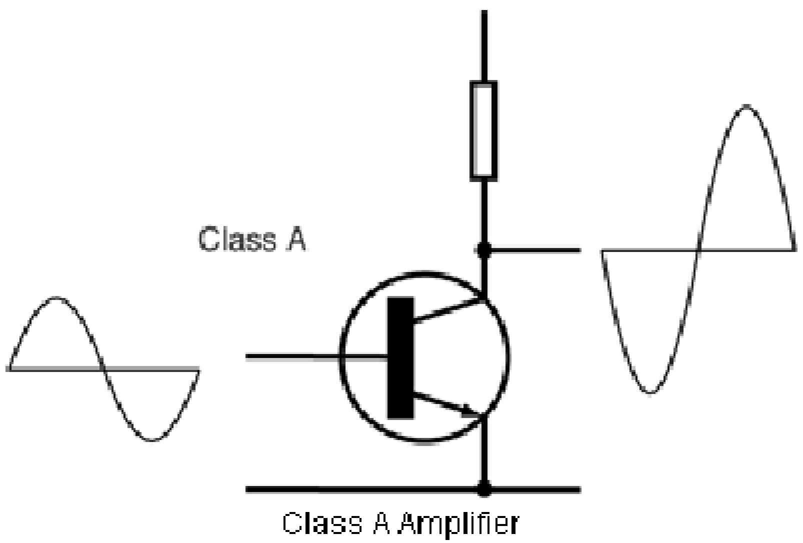
Class-A Benefits

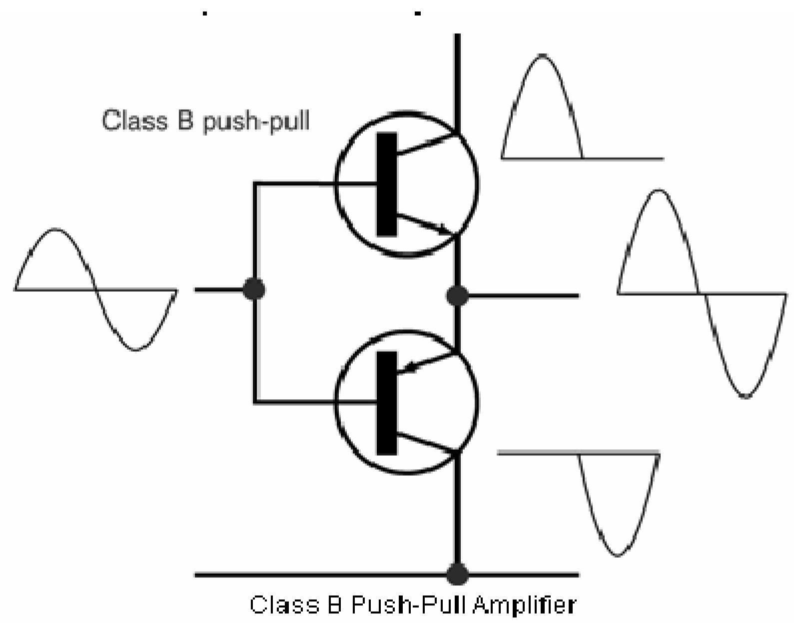
- No cross over distortion
- No switching distortion
- Lower harmonic distortion in the voltage amplifier
- Lower harmonic distortion in the current amplifier
- No signal dependent distortion from the power supply
- Constant and low output impedance
- Simpler design

Power Amplifier Classes – “B”

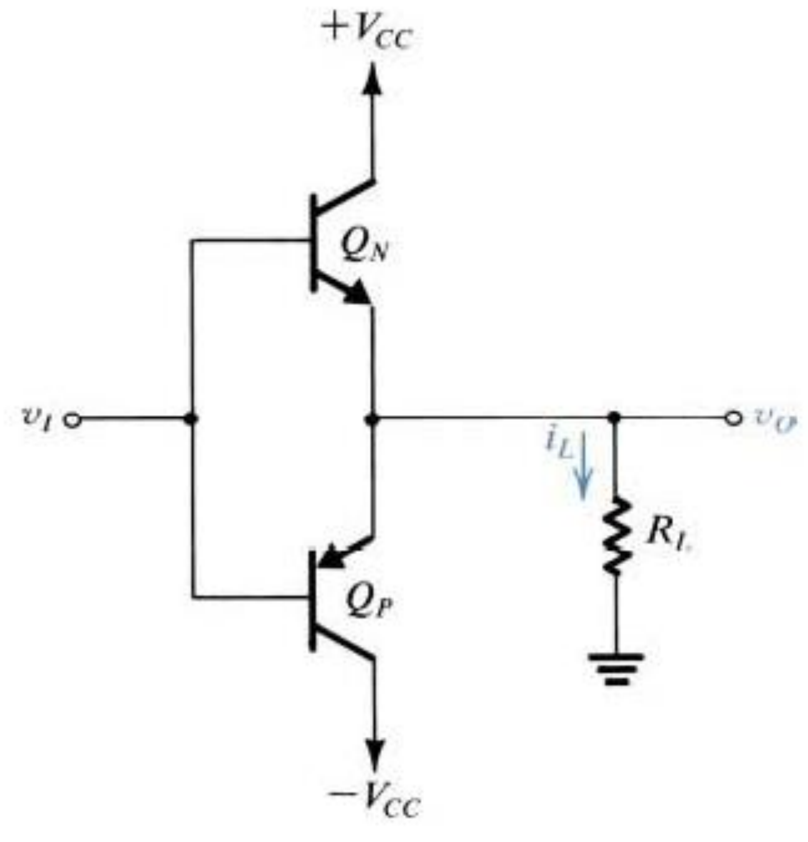
- Class “B”
  - opposite of class A: both output devices are never allowed to be on at the same time
  - each output device is on for exactly one half of a complete sinusoidal signal cycle
  - class B designs show high efficiency but poor linearity around the crossover region (due to the time it takes to turn one device off and the other device on, which translates into extreme crossover distortion)
  - class B designs restricted to low power applications, e.g., battery operated equipment, such as communications audio

Class A vs. Class B



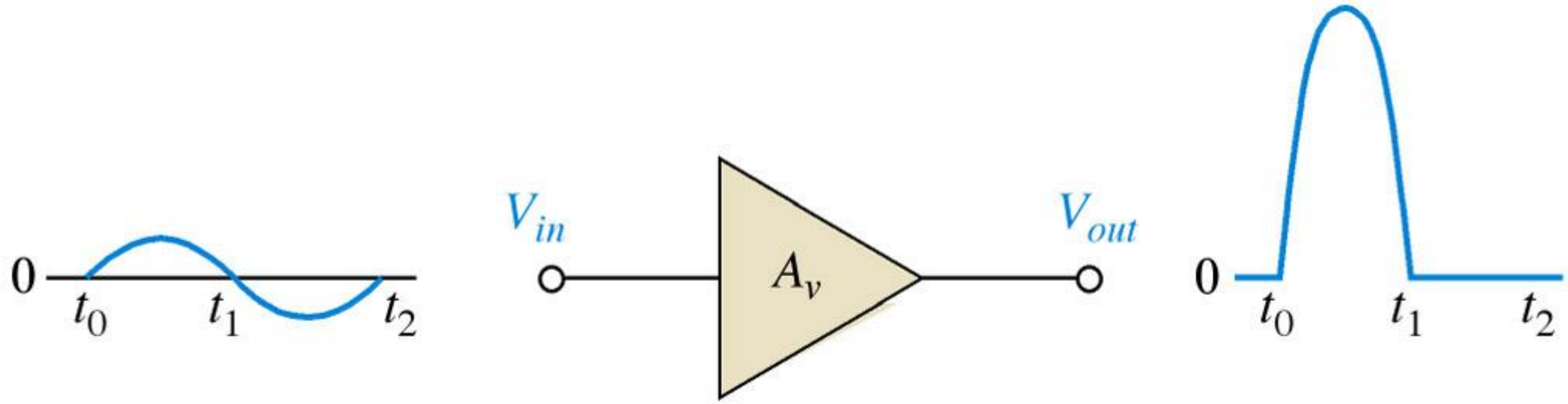


Circuit Operation

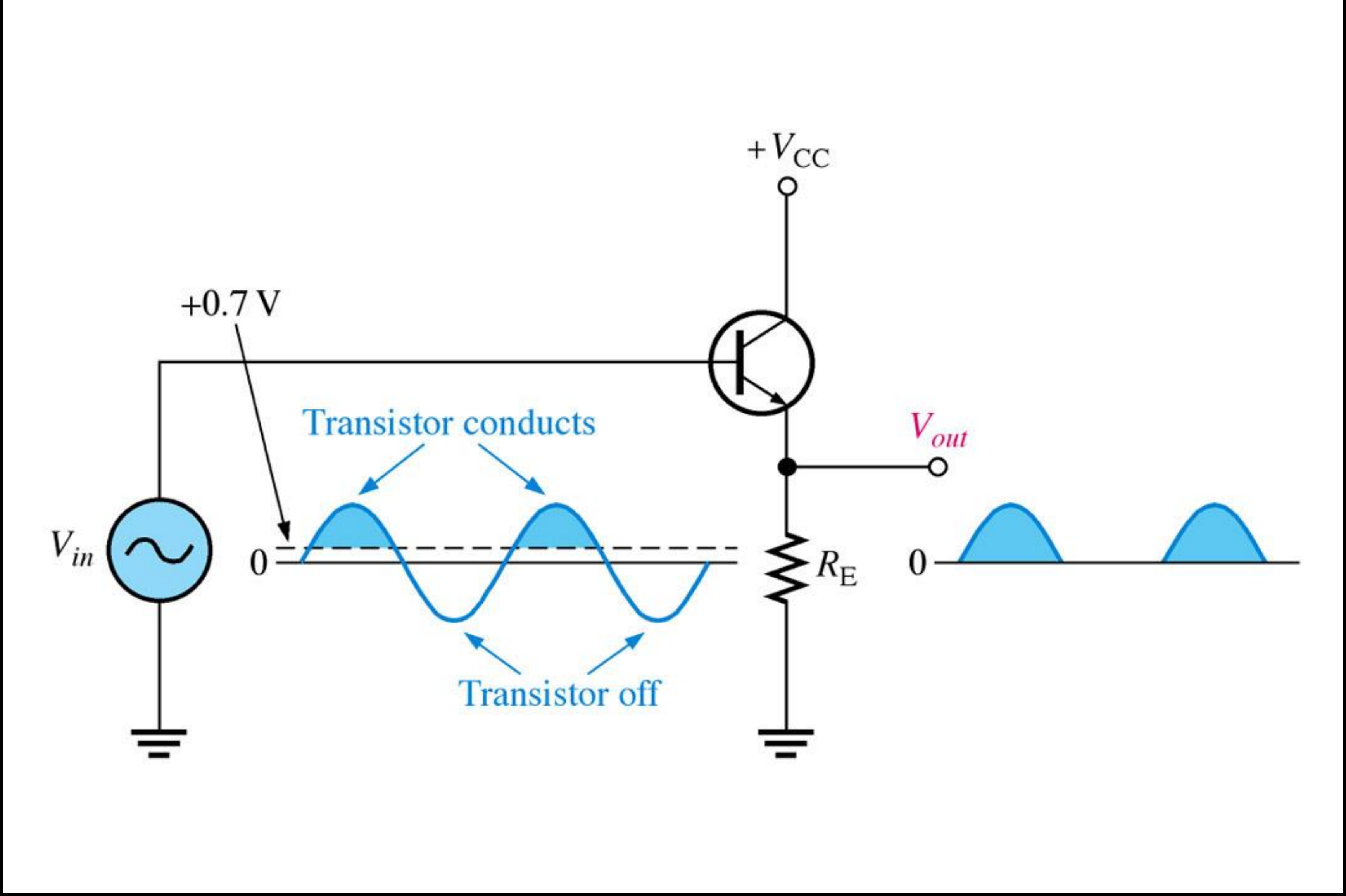


Class B output stage.

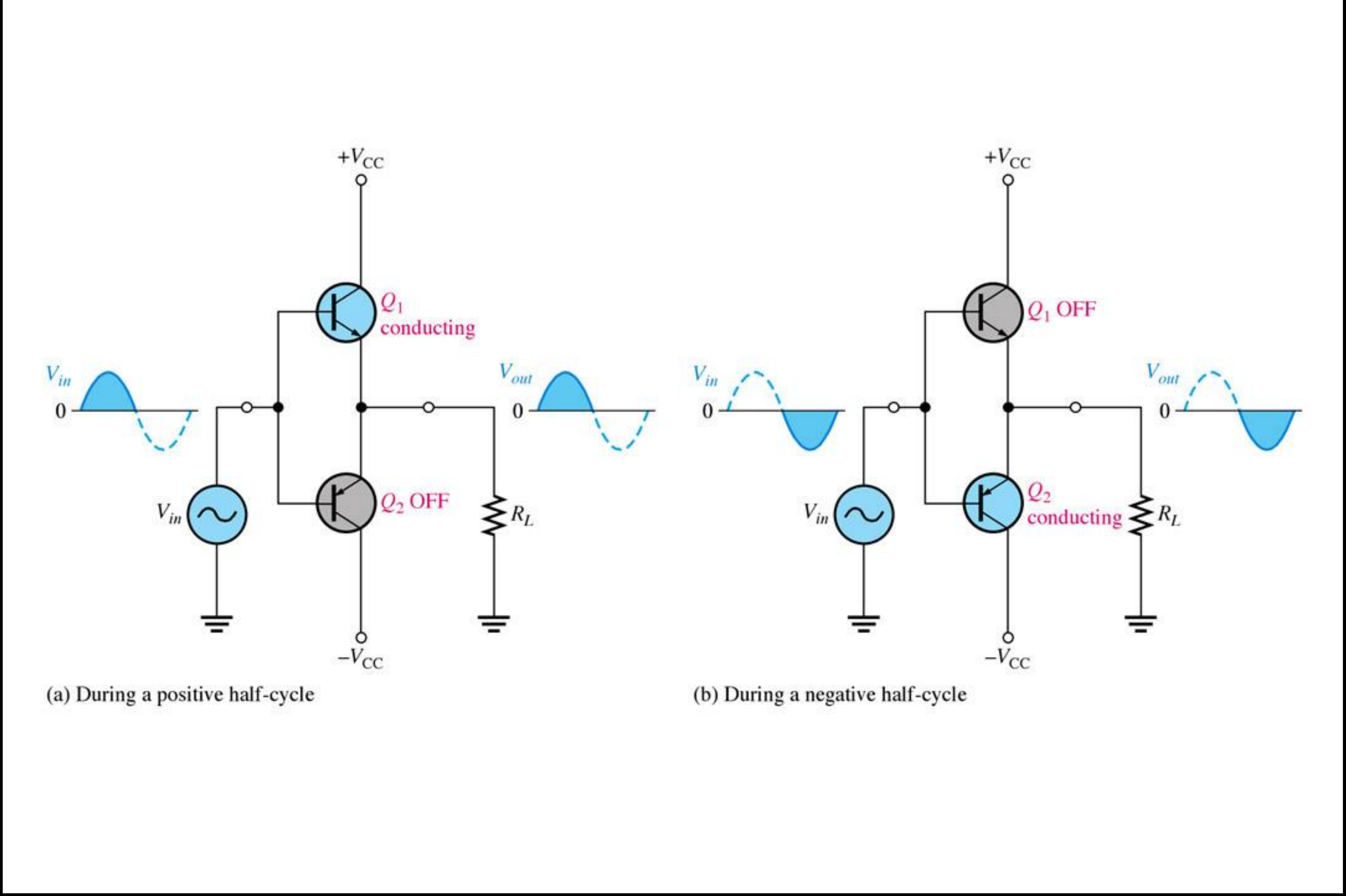
Basic class B amplifier operation (noninverting).



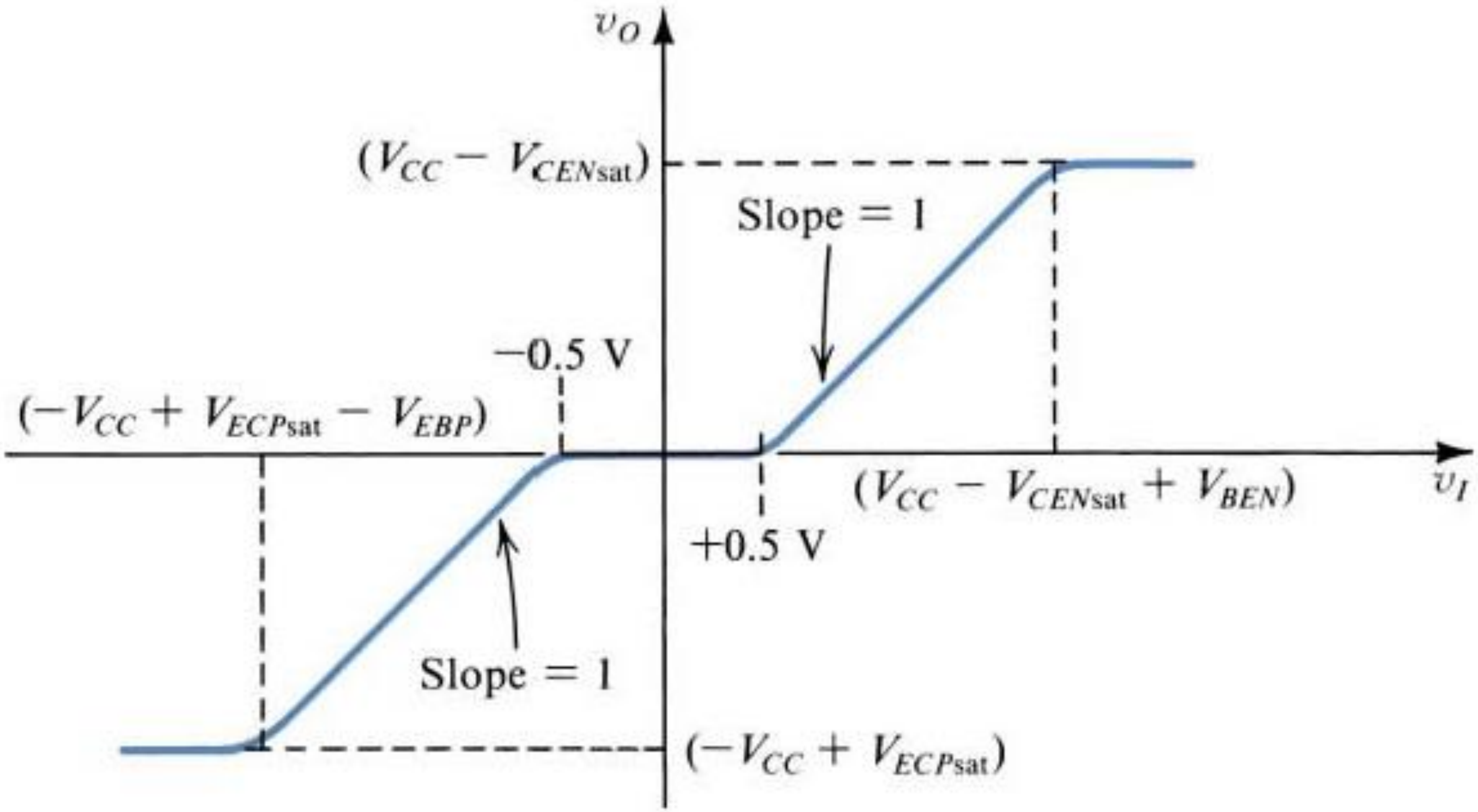
Common-collector class B amplifier.



Class B push-pull ac operation.

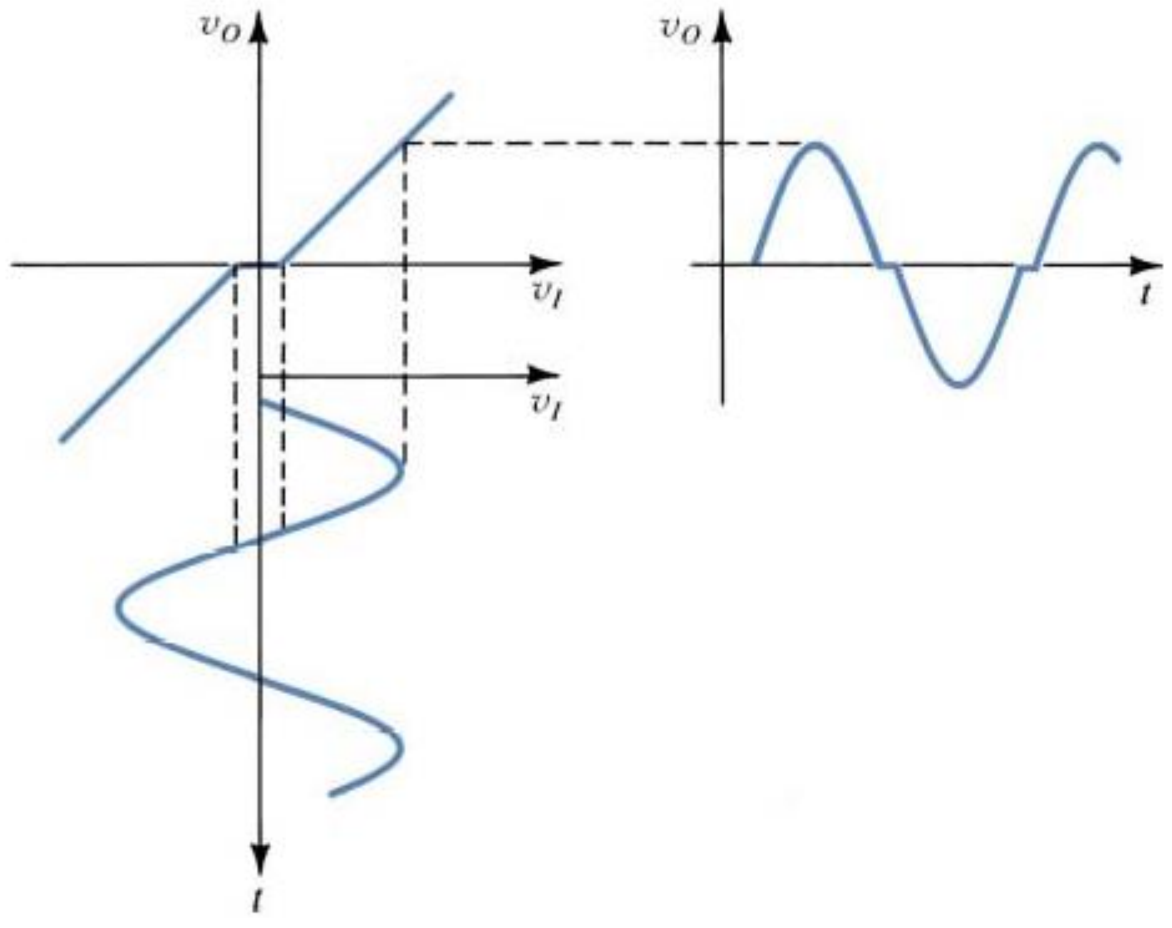


Transfer Characteristics



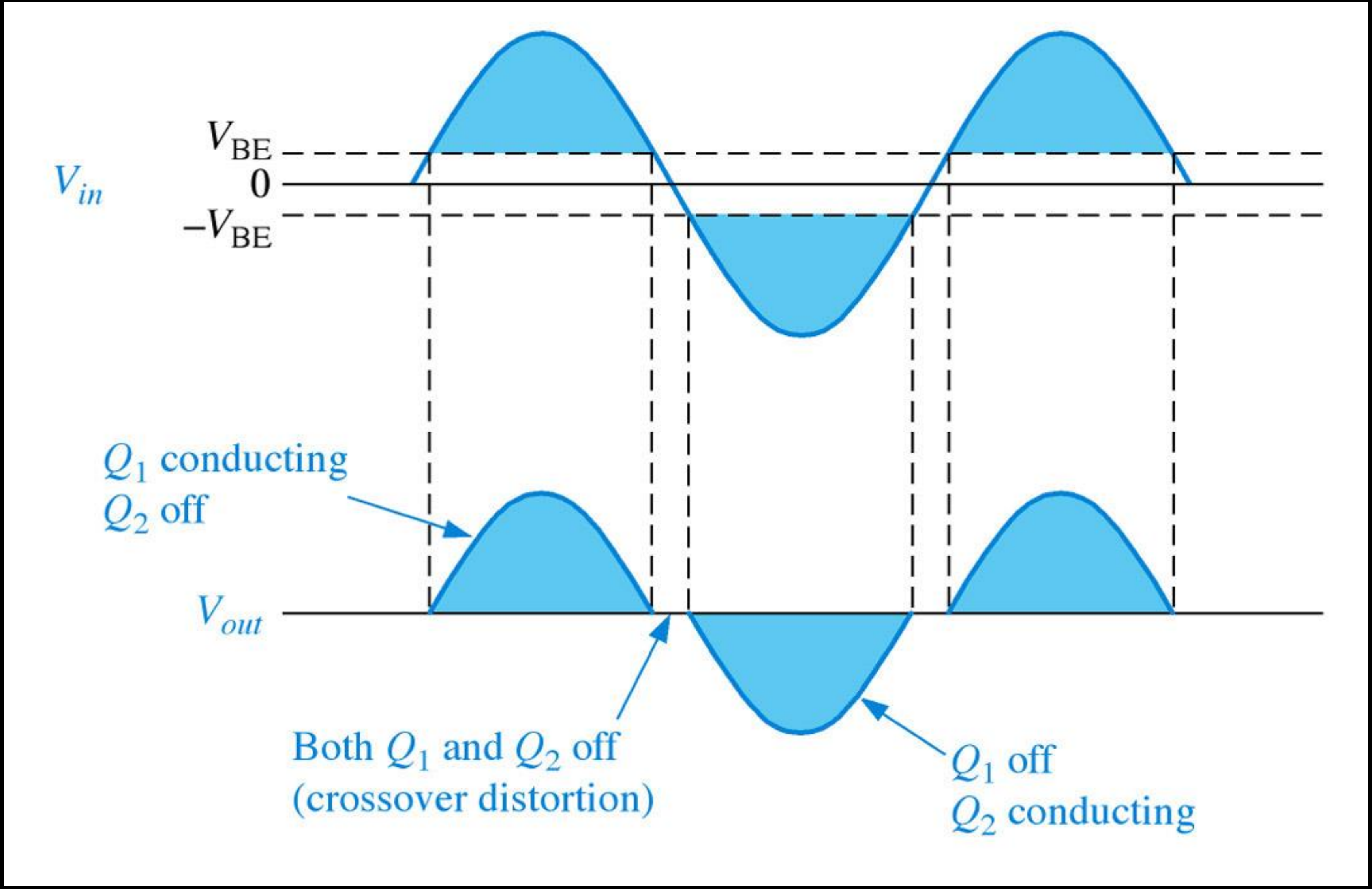
Transfer characteristic for the class B output stage.

Crossover Distortion

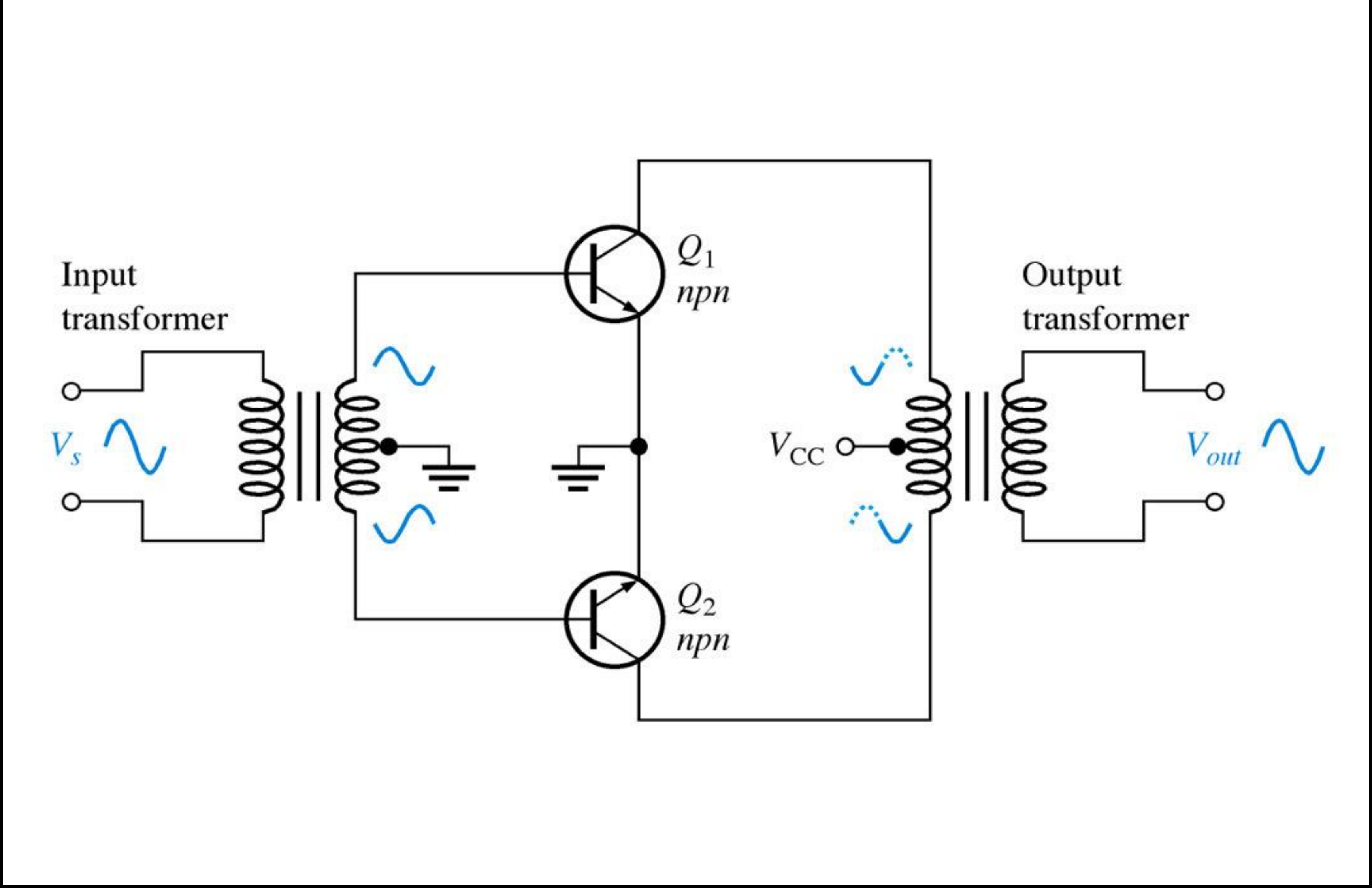


How the dead band in the class B transfer characteristic results in crossover distortion.

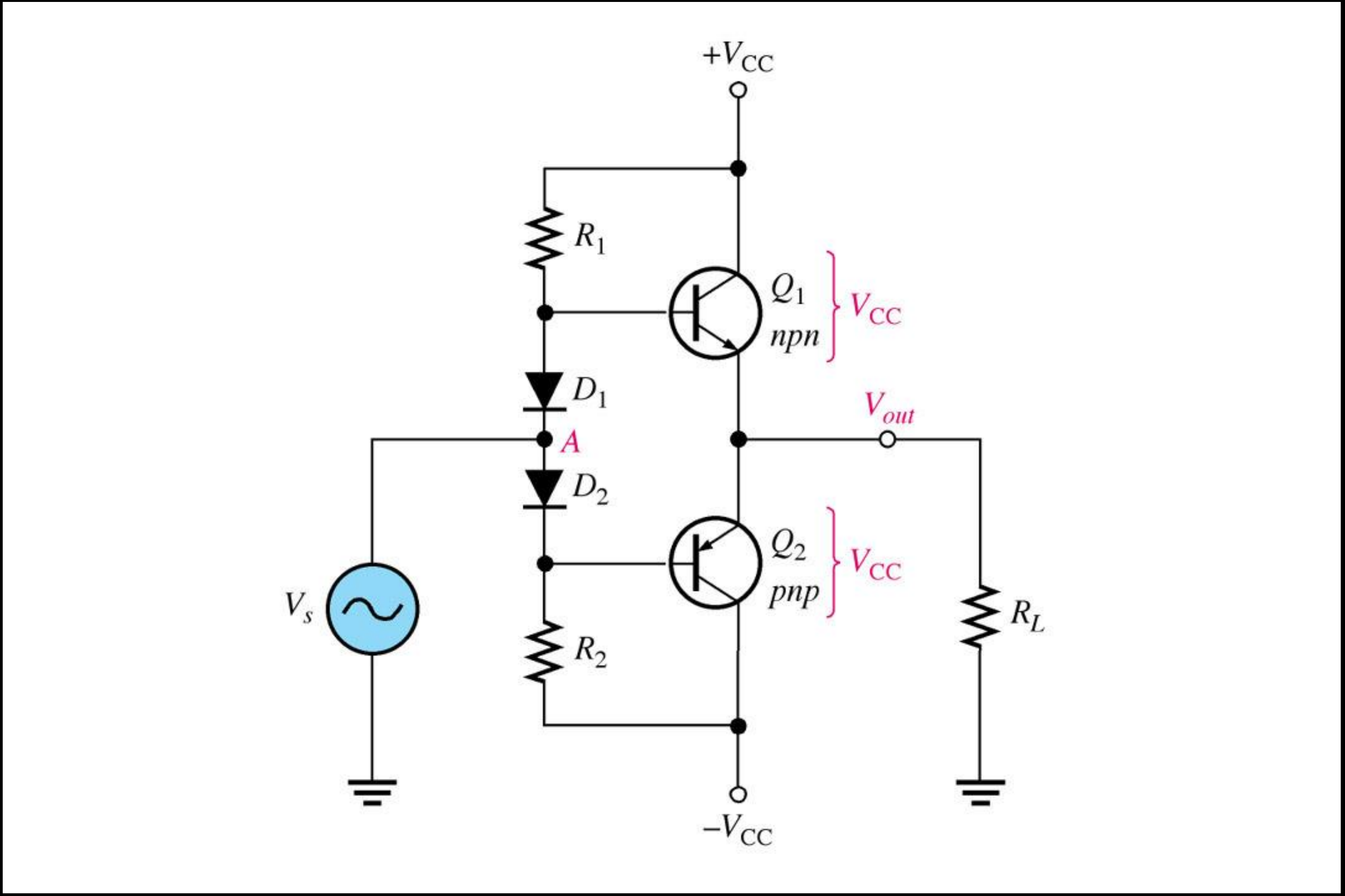
Illustration of crossover distortion in a class B push-pull amplifier. The transistors conduct only during the portions of the input indicated by the shaded areas.

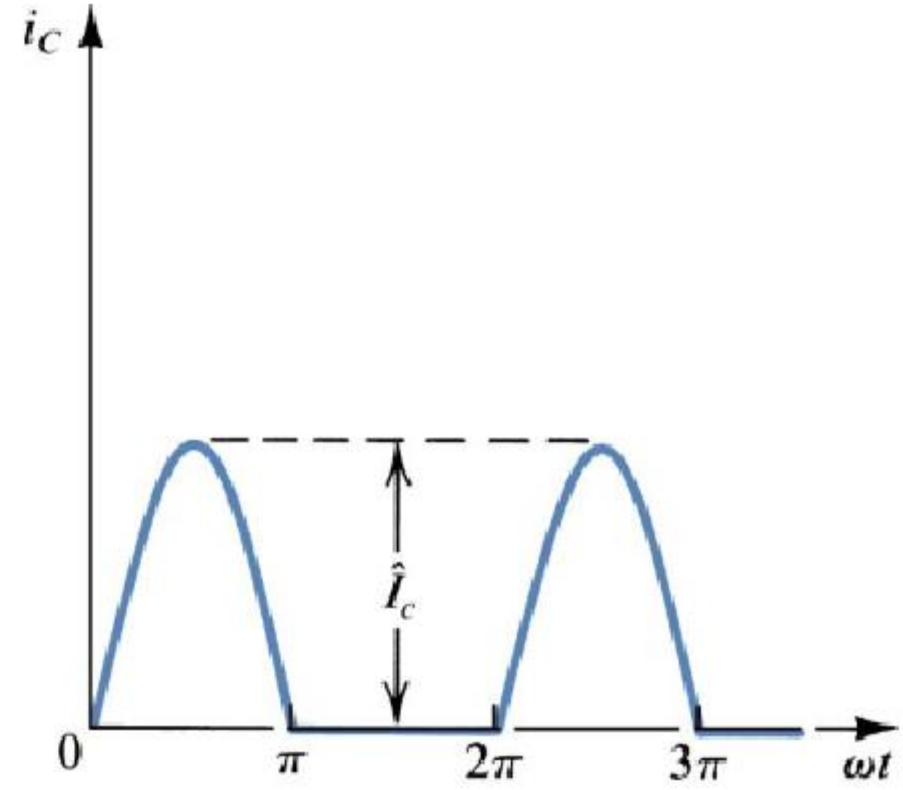


Transformer coupled push-pull amplifiers. Q_1 conducts during the positive half-cycle; Q_2 conducts during the negative half-cycle. The two halves are combined by the output transformer.

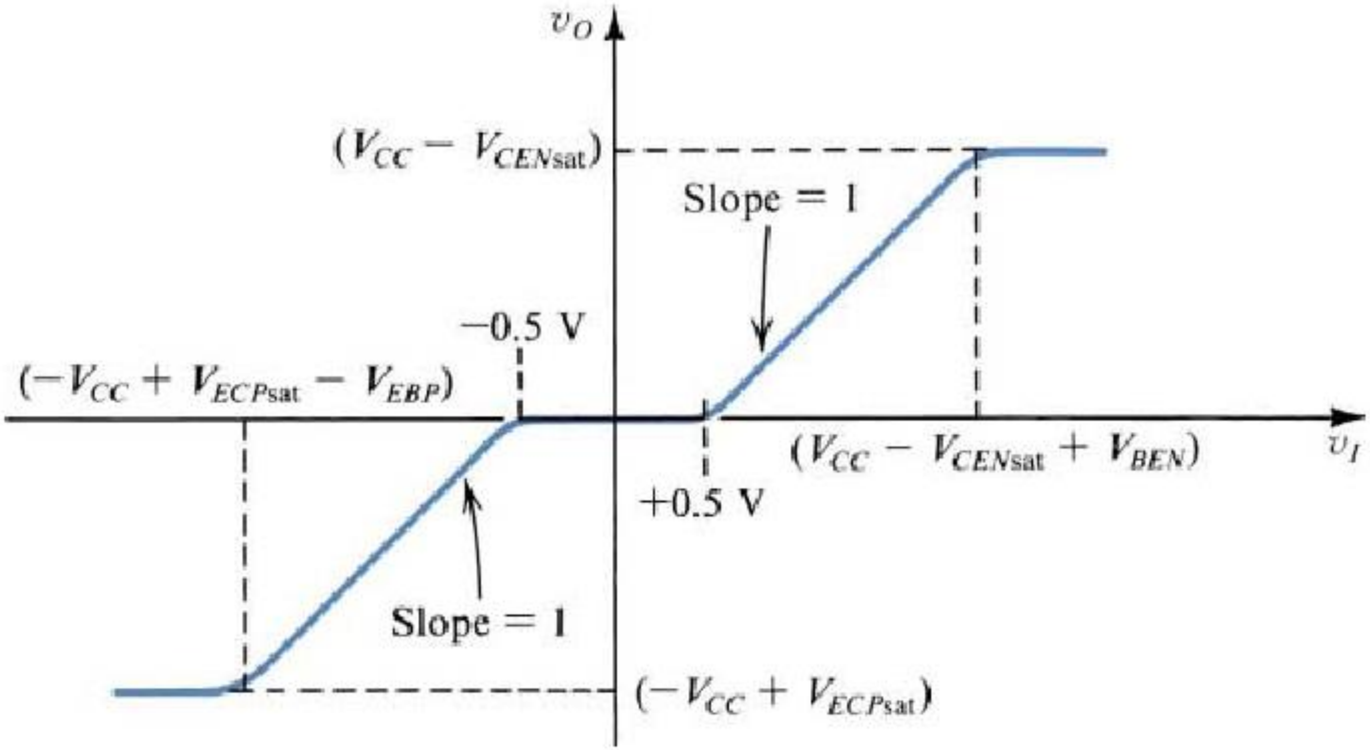


Biasing the push-pull amplifier to eliminate crossover distortion.



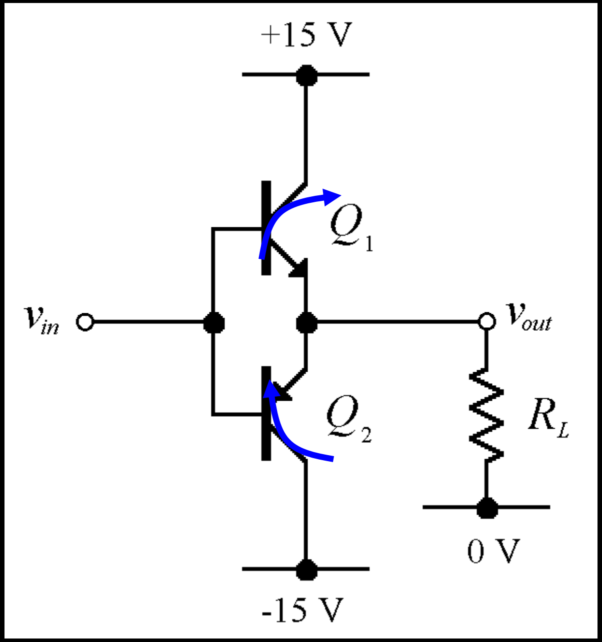


Collector current waveforms for transistors operating in class B amplifier stages



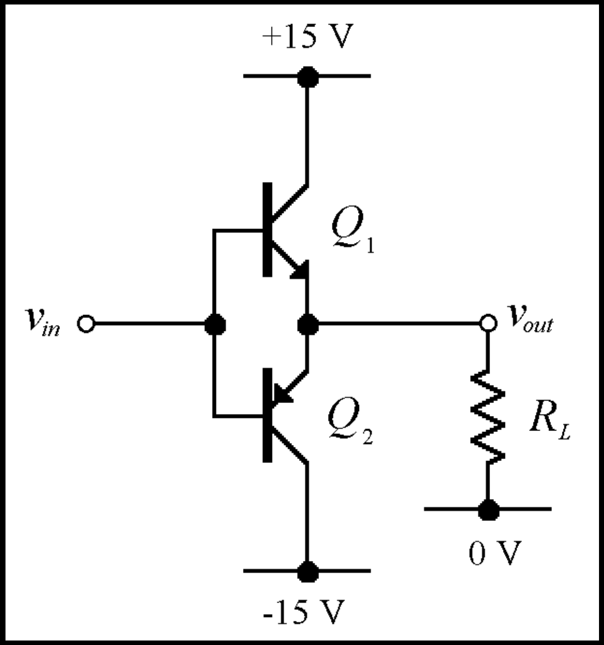
Transfer characteristic for the class B output stage.

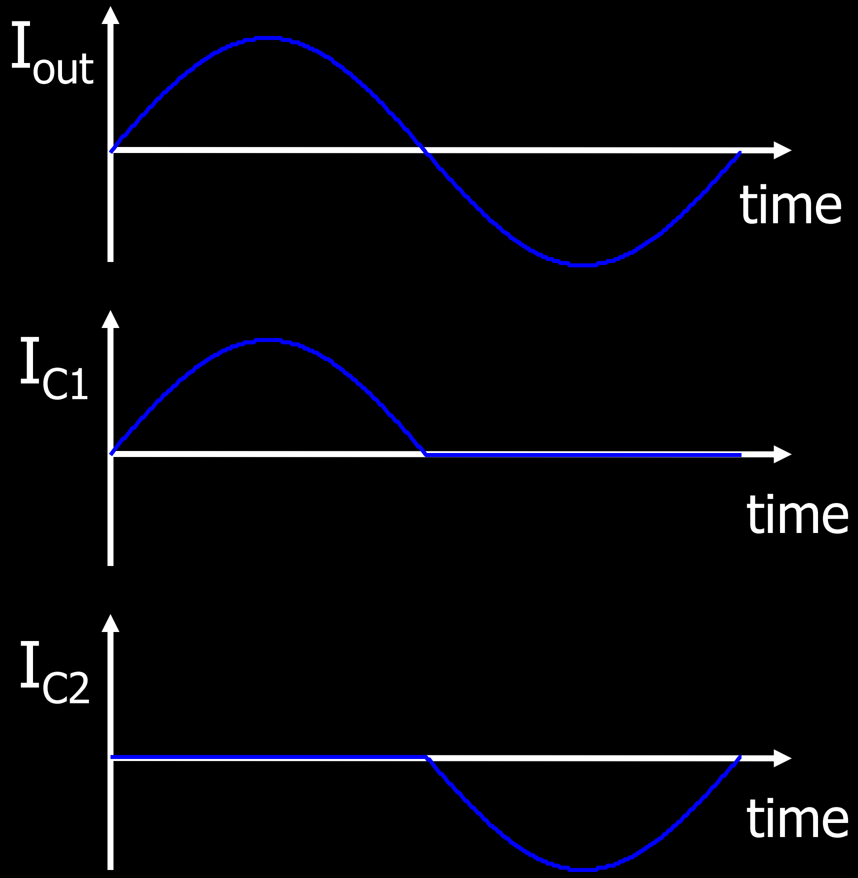
Class B Output Stage



- Q_1 and Q_2 form two unbiased emitter followers
  - Q_1 only conducts when the input is positive
  - Q_2 only conducts when the input is negative
- Conduction angle is, therefore, 180°
- When the input is zero, neither conducts
- i.e. the quiescent power dissipation is zero

Class B Current Waveforms





Maximum Output Power

$$P_{out} = I_{out(rms)}V_{out(rms)}$$

$$I_{out(rms)} = 0.707I_{out(peak)} = 0.707I_{c(sat)}$$

$$V_{out(rms)} = 0.707V_{out(peak)} = 0.707V_{CEQ}$$

$$P_{out} = 0.5I_{c(sat)}V_{CEQ}$$

Substituting $V_{CC}/2$ for V_{CEQ}, the maximum average output power is

$$\mathbf{P_{out} = 0.25I_{c(sat)}V_{CC}}$$

DC Input Power

The dc input power comes from the V_{CC} supply and is

$$P_{DC} = I_{CC}V_{CC}$$

Since each transistor draws current for a half-cycle, the current is a half-wave signal with an average value of

$$I_{CC} = \frac{I_{c(sat)}}{\pi}$$

$$P_{DC} = \frac{I_{c(sat)}V_{CC}}{\pi}$$

Efficiency

$$\eta = \frac{P_{out}}{P_{DC}}$$

$$\eta_{max} = \frac{P_{out}}{P_{DC}} = \frac{0.25I_{c(sat)}V_{CC}}{I_{c(sat)}V_{CC}/\pi} = 0.25\pi$$

$$\boldsymbol{\eta_{max} = 0.79}$$

Class B

- A class B output stage can be far more efficient than a class A stage (78.5 % maximum efficiency compared with 25 %).
- It also requires twice as many output transistors…
- …and it isn't very linear; cross-over distortion can be significant.

Class B

- Class B amplifiers are used in low cost designs or designs where sound quality is not that important.
- Class B amplifiers are significantly more efficient than class A amps.
- They suffer from bad distortion when the signal level is low (the distortion in this region of operation is called "crossover distortion").

Class B

- Class B is used most often where economy of design is needed.

- Before the advent of IC amplifiers, class B amplifiers were common in clock radio circuits, pocket transistor radios, or other applications where quality of sound is not that critical.

Power Amplifier Classes – "AB"

- Class "AB"
  - intermediate case: both devices are allowed to be on at the same time, *but just barely*
  - output bias is set so that current flows in a specific output device appreciably more than a half cycle but less than the entire cycle (enough to keep each device operating *so they respond instantly* to input voltage demands)
  - the inherent non-linearity of class B designs is eliminated, without the gross inefficiencies of the class A design
  - combination of good efficiency (around 50%) with excellent linearity that makes class AB the most popular consumer audio amplifier design

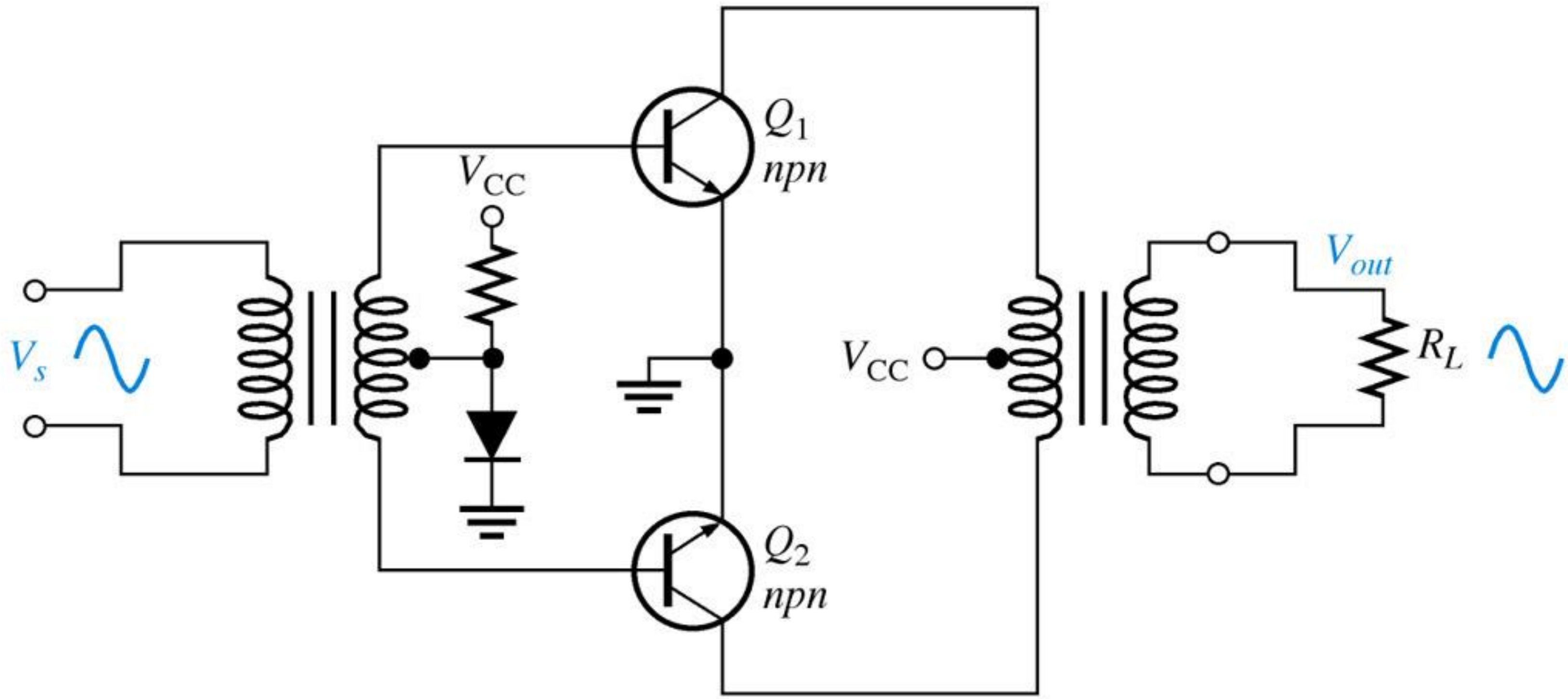
Class AB

- Class AB is probably the most common amplifier class currently used in home stereo and similar amplifiers.

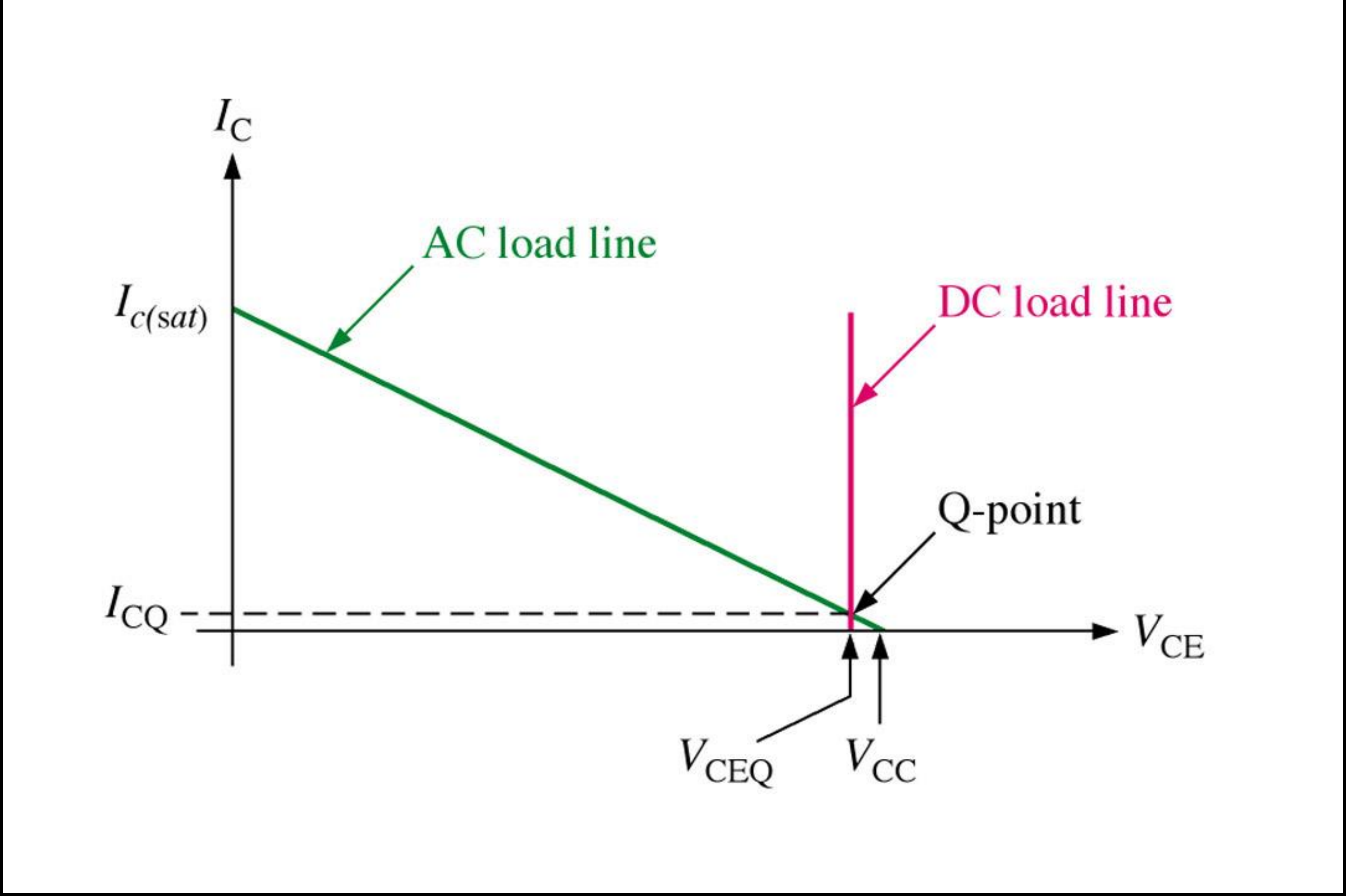
- Class AB amps combine the good points of class A and B amps.

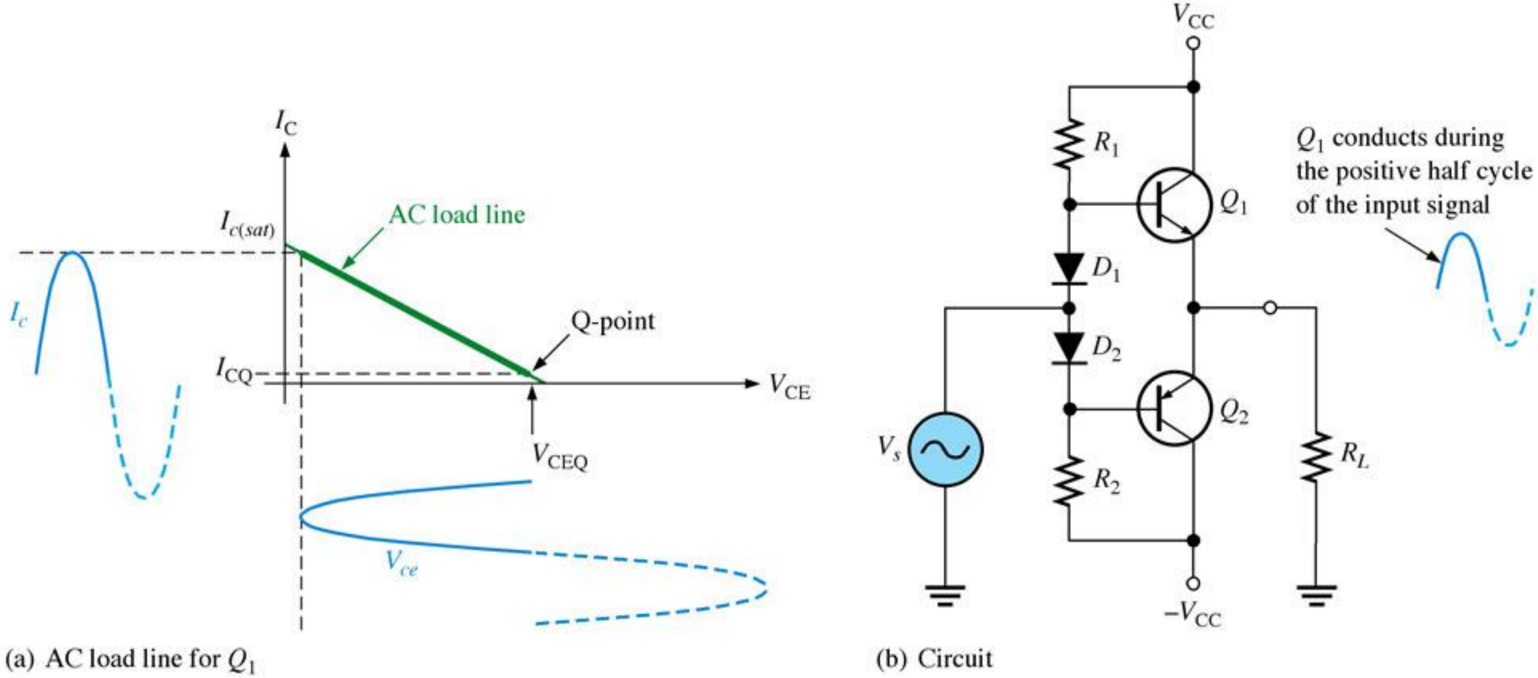
- They have the improved efficiency of class B amps and distortion performance that is a lot closer to that of a class A amp.

Eliminating crossover distortion in a transformer-coupled push-pull amplifier. The diode compensates for the base-emitter drop of the transistors and produces class AB operation.

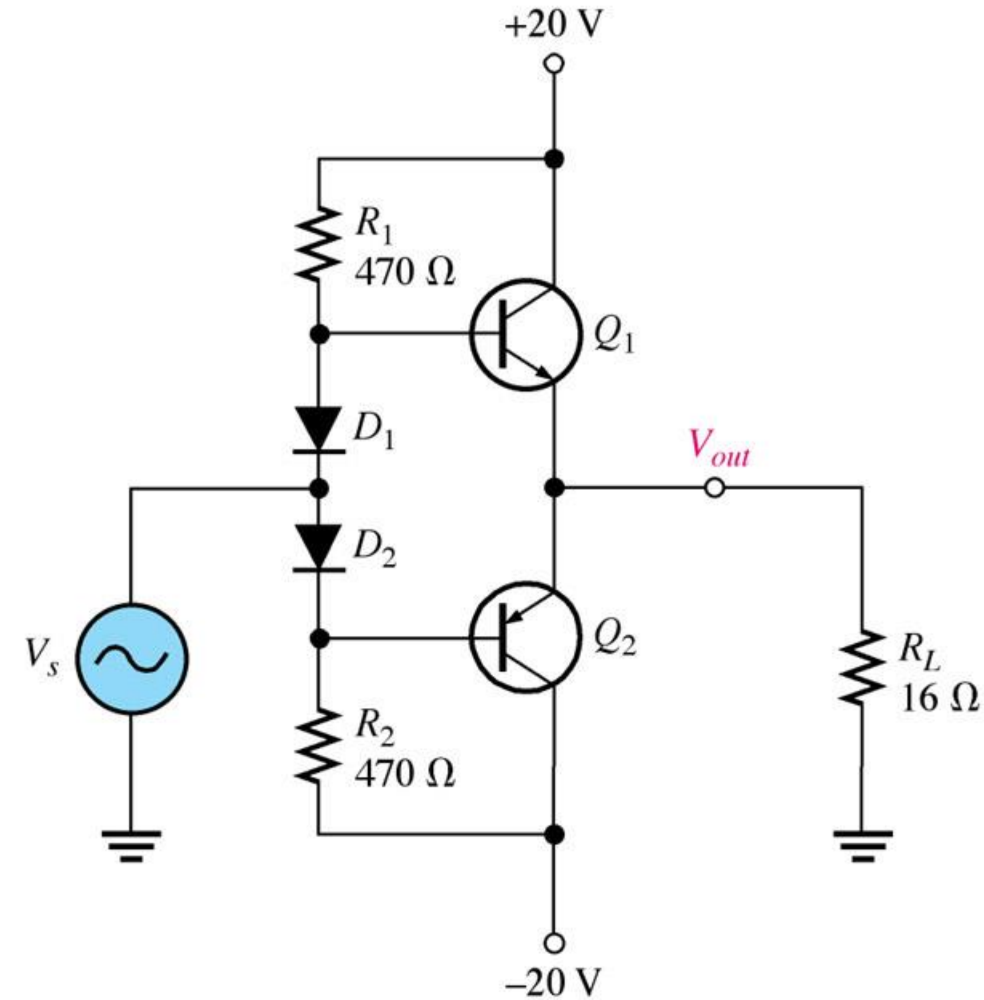


Load lines for a complementary symmetry push-pull amplifier. Only the load lines for the *npn* transistor are shown.

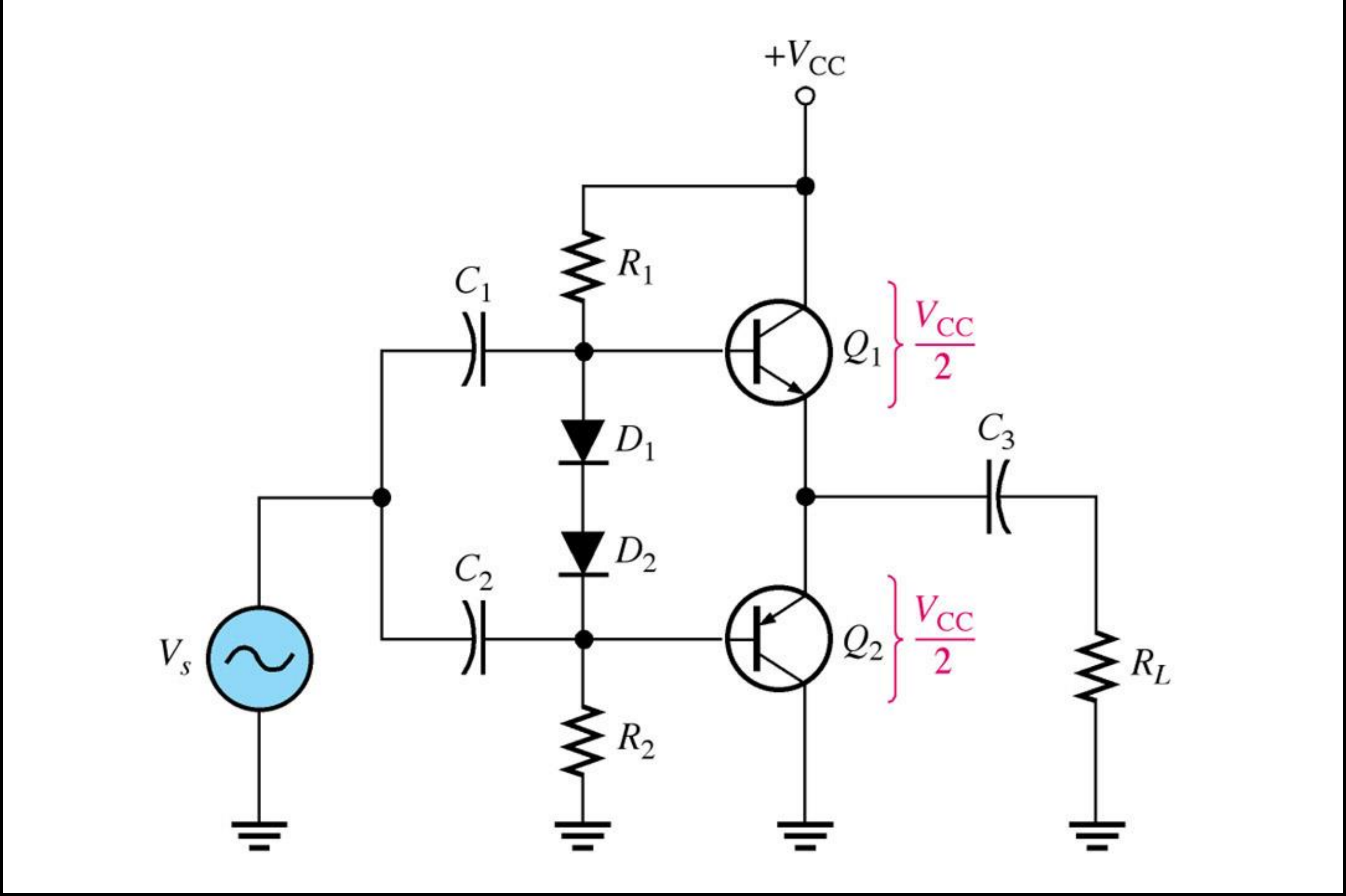


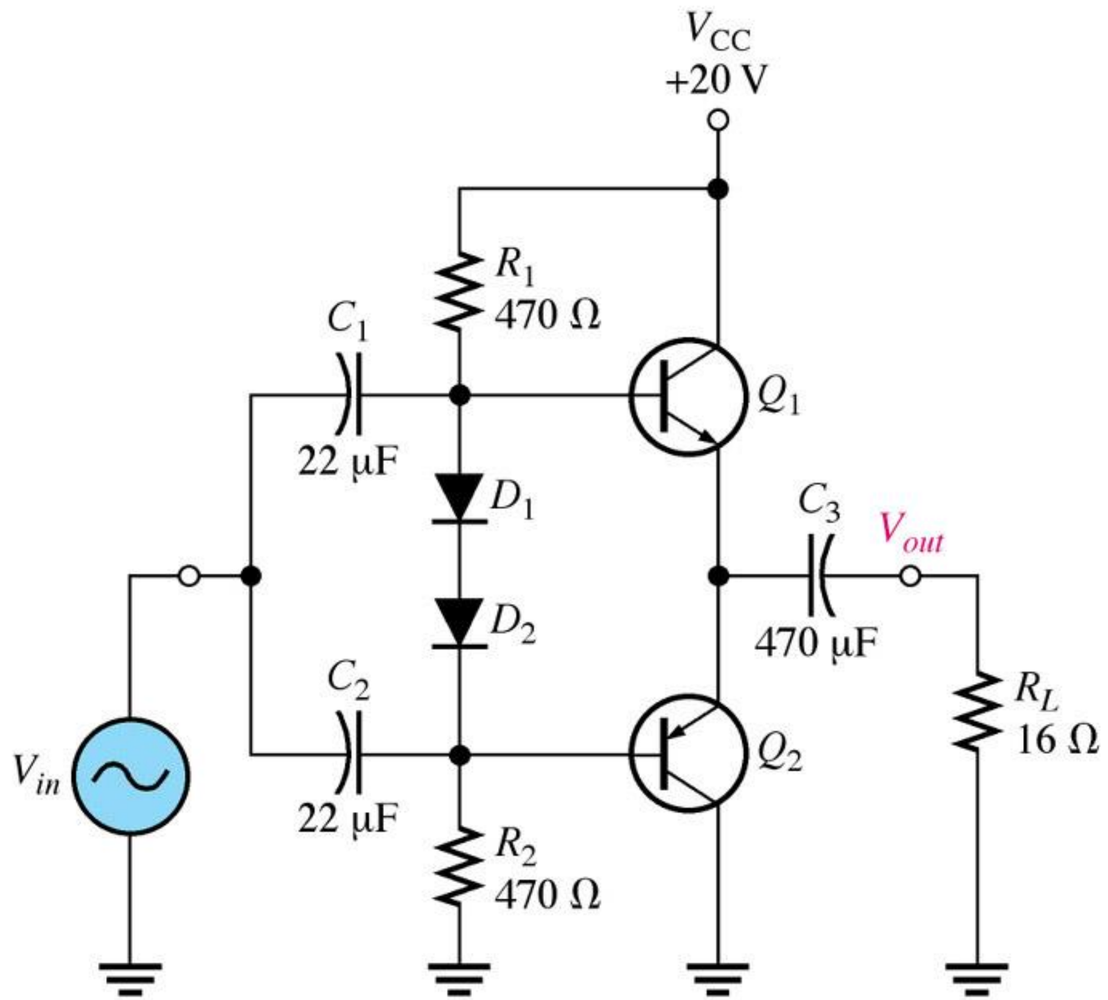


I_C
$I_{c(sat)}$
AC load line
Q-point
I_{CQ}
V_{CE}
V_{CEQ}
I_c
V_{ce}
(a) AC load line for Q_1
V_{CC}
R_1
Q_1
D_1
D_2
Q_2
V_s
R_2
R_L
$-V_{CC}$
Q_1 conducts during the positive half cycle of the input signal
(b) Circuit

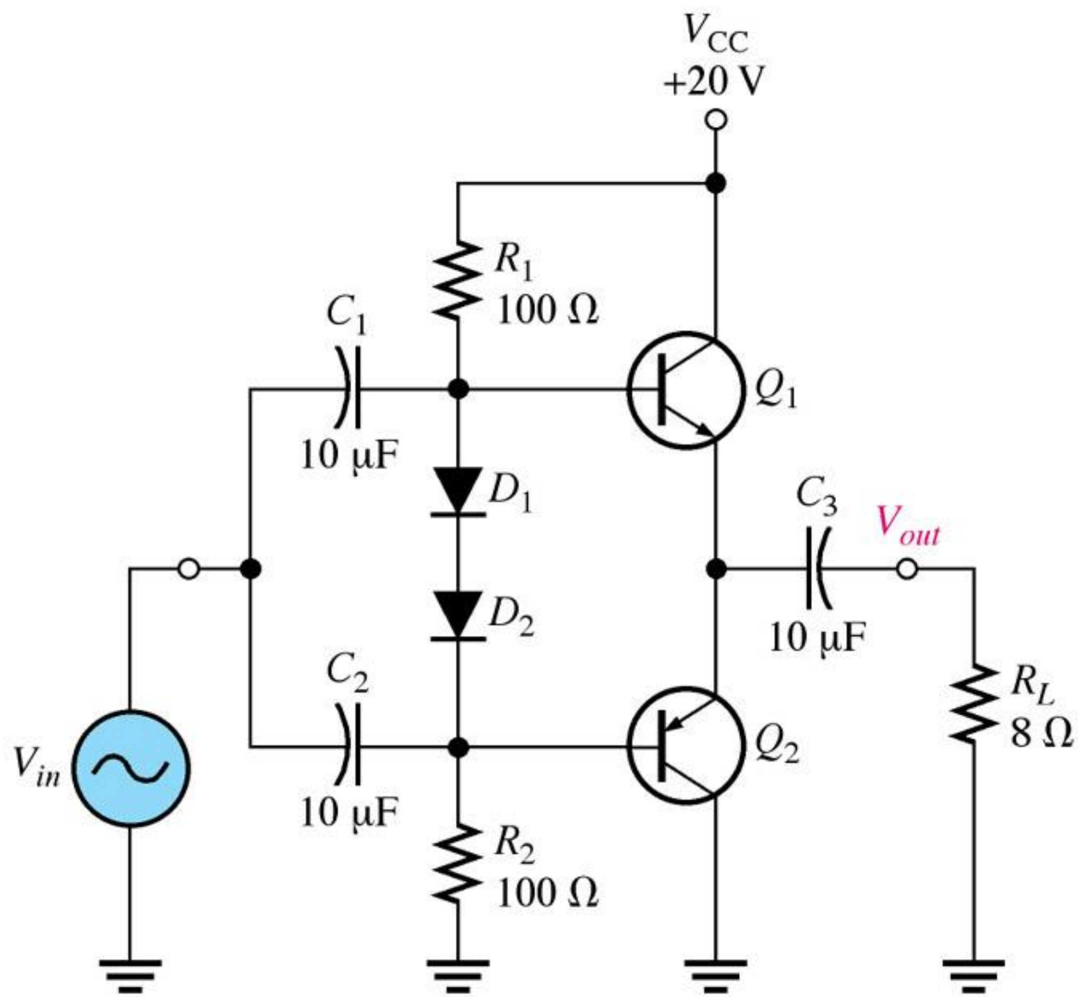

+20 V
R1
470 Ω
Q1
D1
Vout
D2
Q2
Vs
RL
16 Ω
R2
470 Ω
–20 V

Single-ended push-pull amplifier.



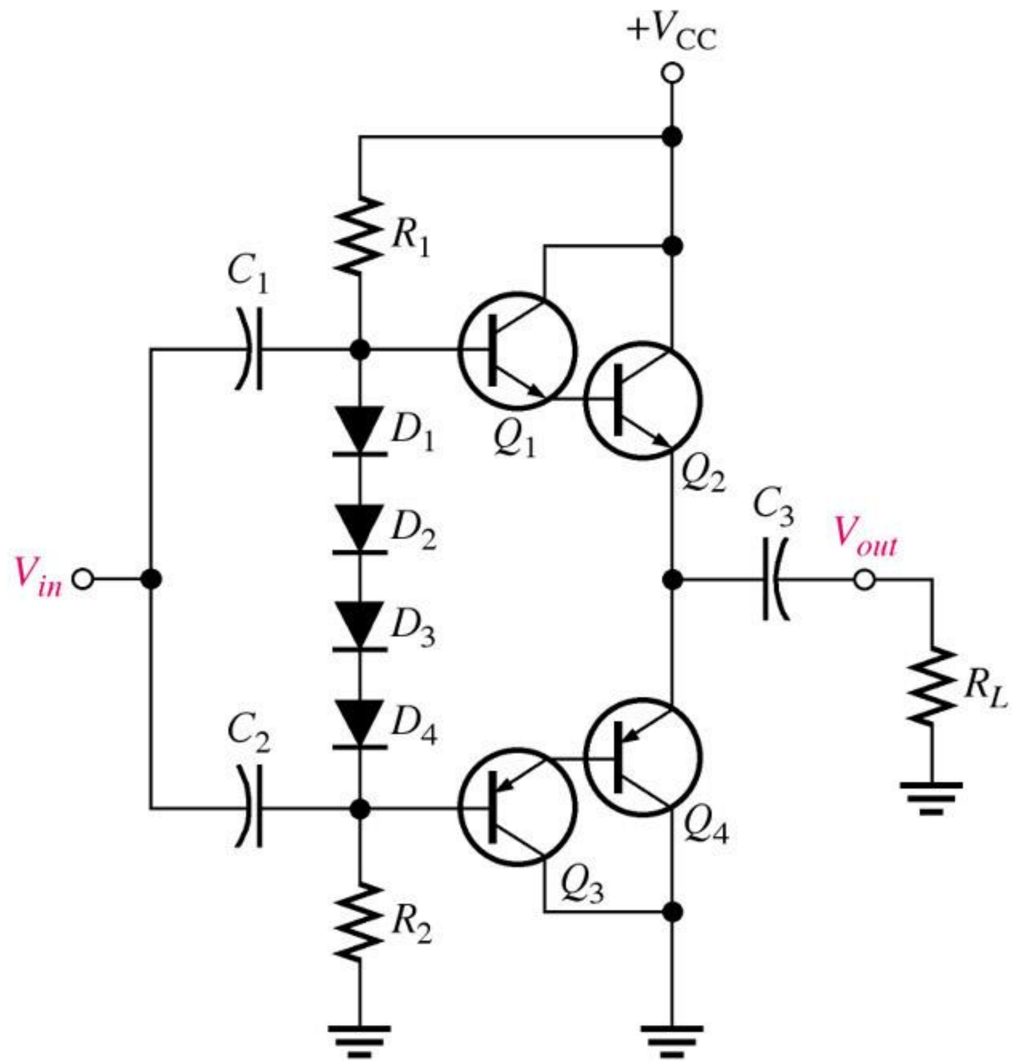


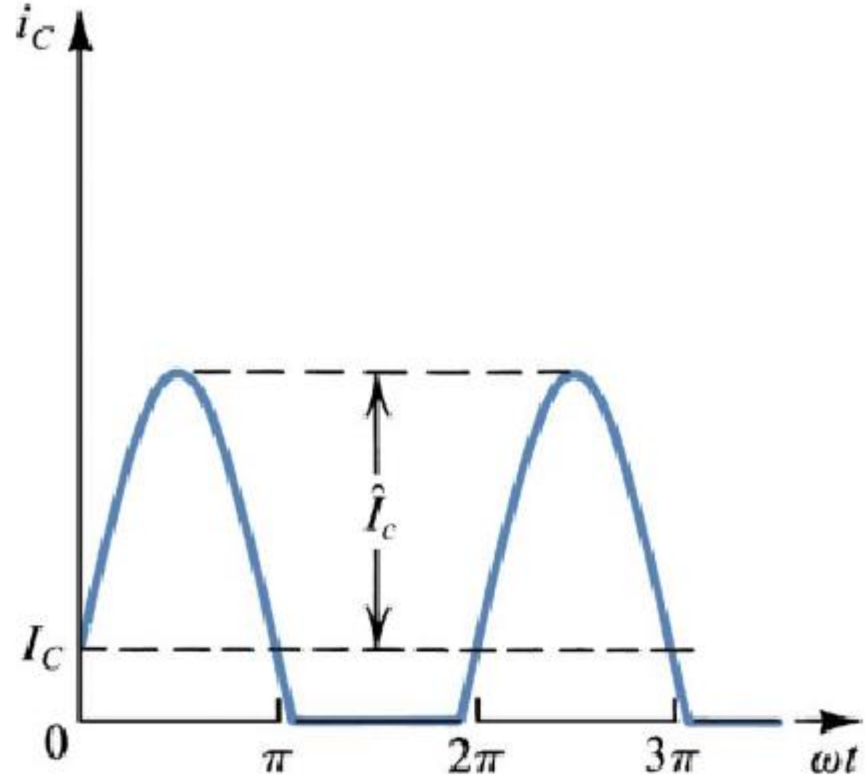
V_CC
+20 V
R_1
470 Ω
C_1
22 μF
Q_1
D_1
C_3
V_out
D_2
470 μF
C_2
R_L
16 Ω
Q_2
V_in
22 μF
R_2
470 Ω



VCC
+20 V
R1
100 Ω
C1
10 μF
Q1
D1
C3
Vout
D2
10 μF
C2
RL
8 Ω
Vin
Q2
10 μF
R2
100 Ω

A Darlington class AB push-pull amplifier.





Collector current waveforms for transistors operating in class AB amplifier stage

FIGURE 9-32 A Class AB push-pull amplifier with correct output voltage.

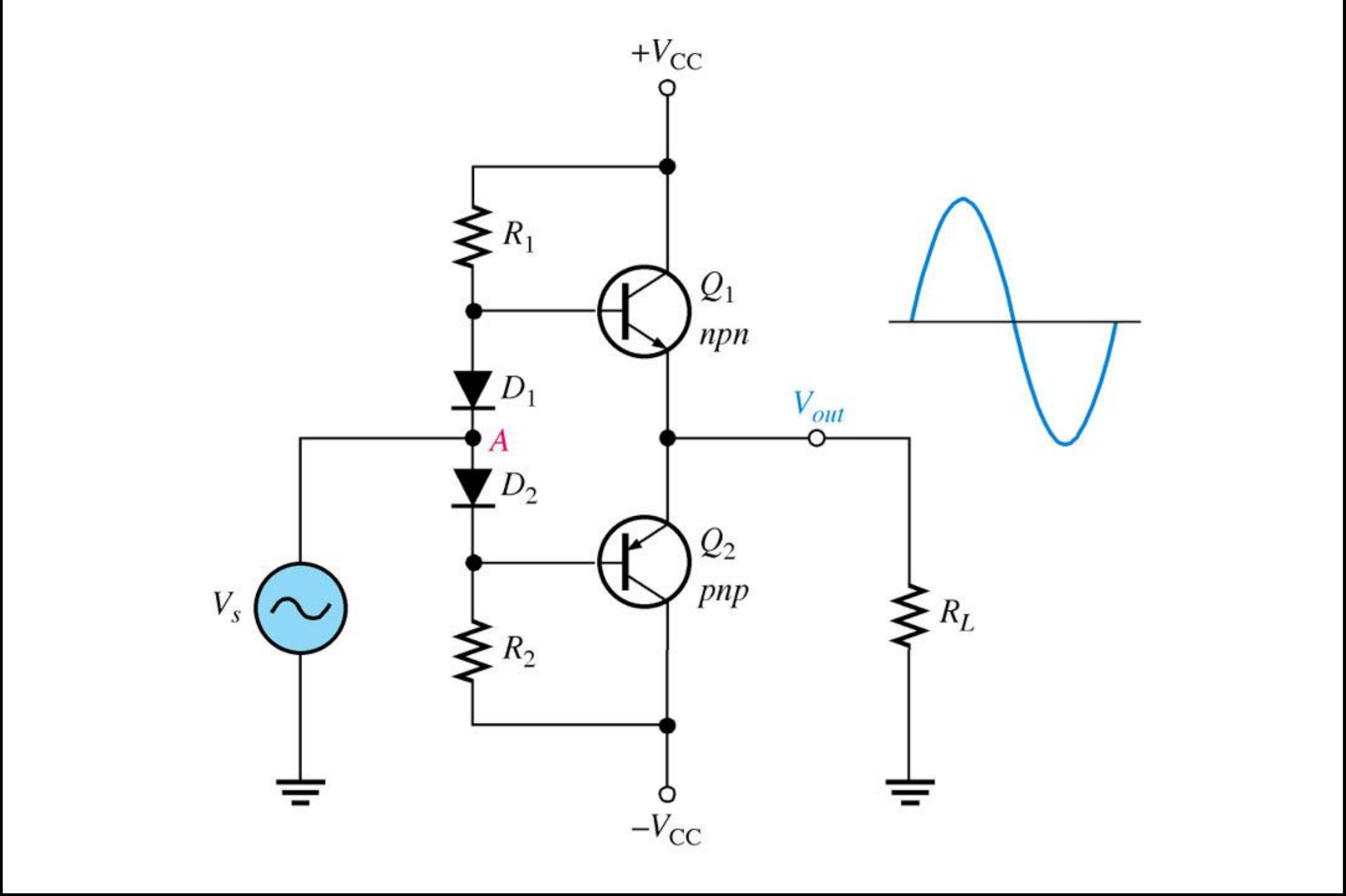
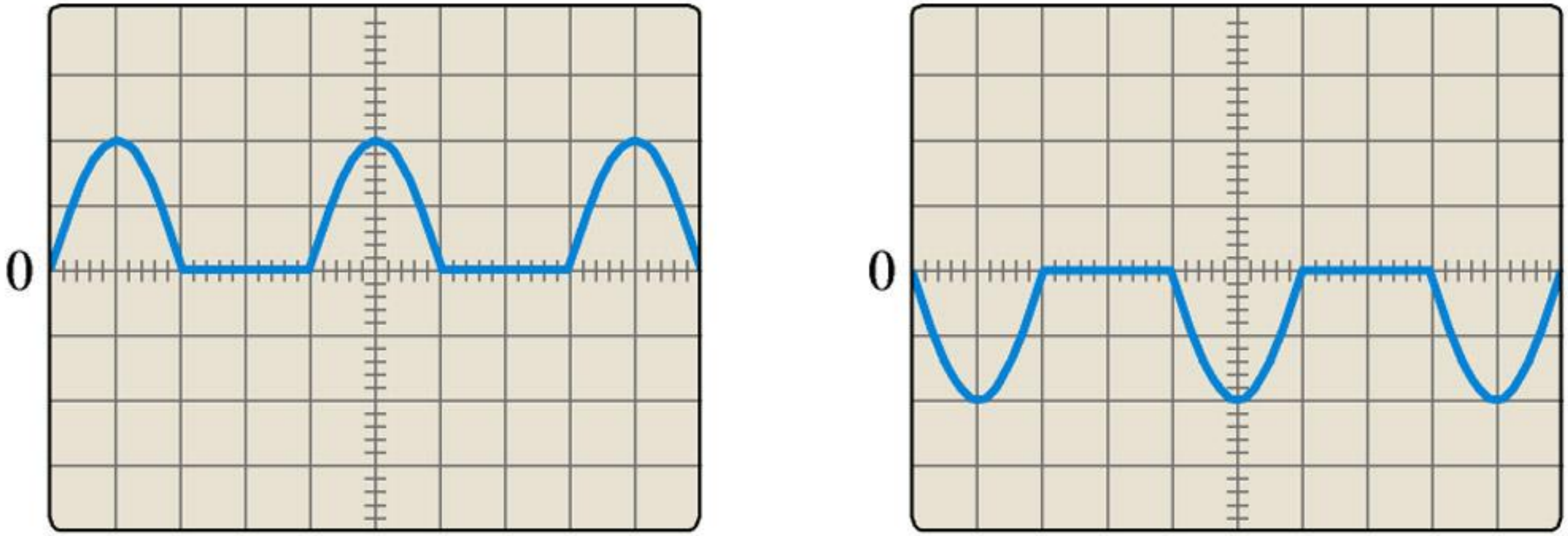


FIGURE 9-33 Incorrect output waveforms for the amplifier in Figure 9-32.



(a) D_1 open or Q_2 base-emitter open

(b) D_2 open or Q_1 base-emitter open

Class AB

- With such amplifiers, distortion is worst when the signal is low, and generally lowest when the signal is just reaching the point of clipping.

- Class AB amps use pairs of transistors, both of them being biased slightly ON so that the crossover distortion (associated with Class B amps) is largely eliminated.

Class C

- Class C amps are never used for audio circuits.

- They are commonly used in RF circuits.

- Class C amplifiers operate the output transistor in a state that results in tremendous distortion (it would be totally unsuitable for audio reproduction).

FIGURE 9-22 Basic class C amplifier operation (non inverting).

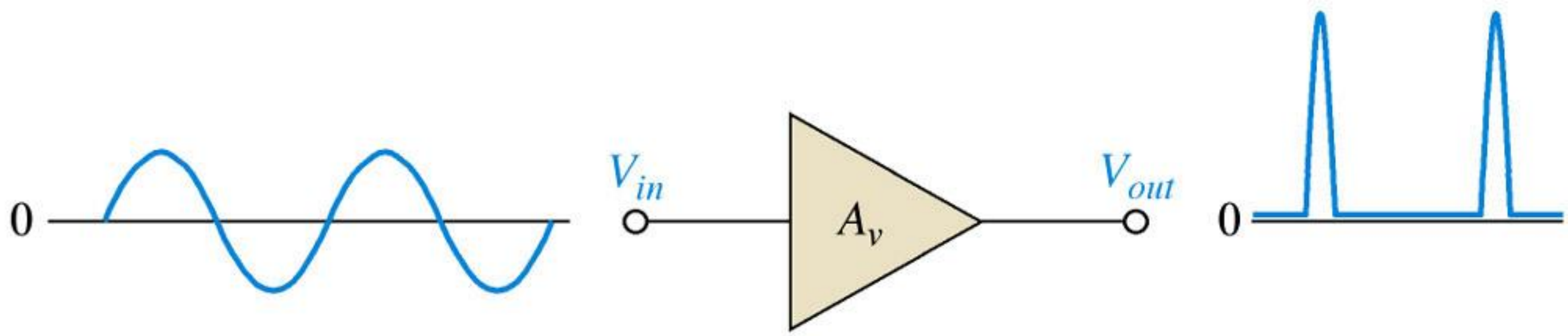


FIGURE 9-23 Basic class C operation.

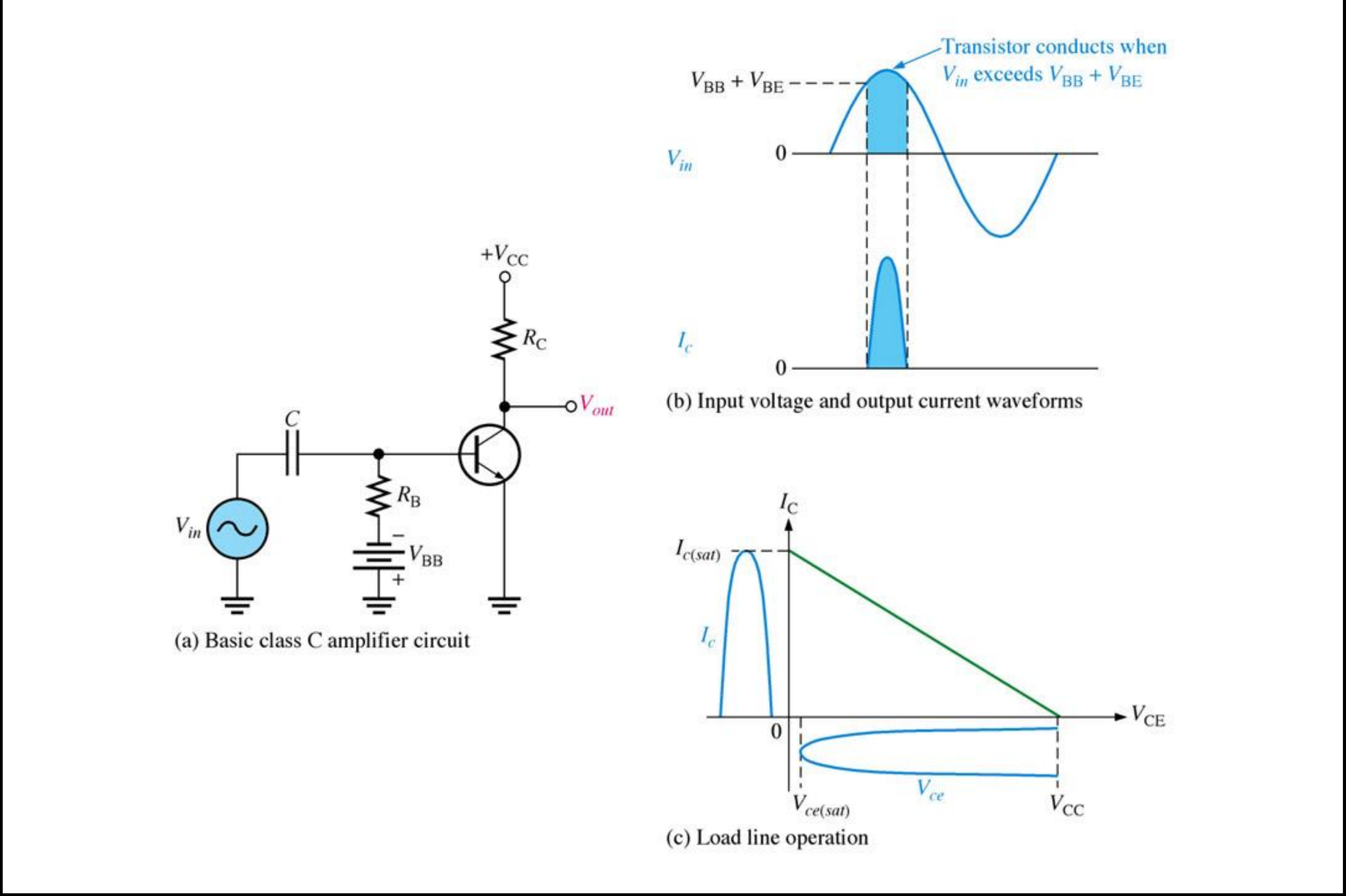
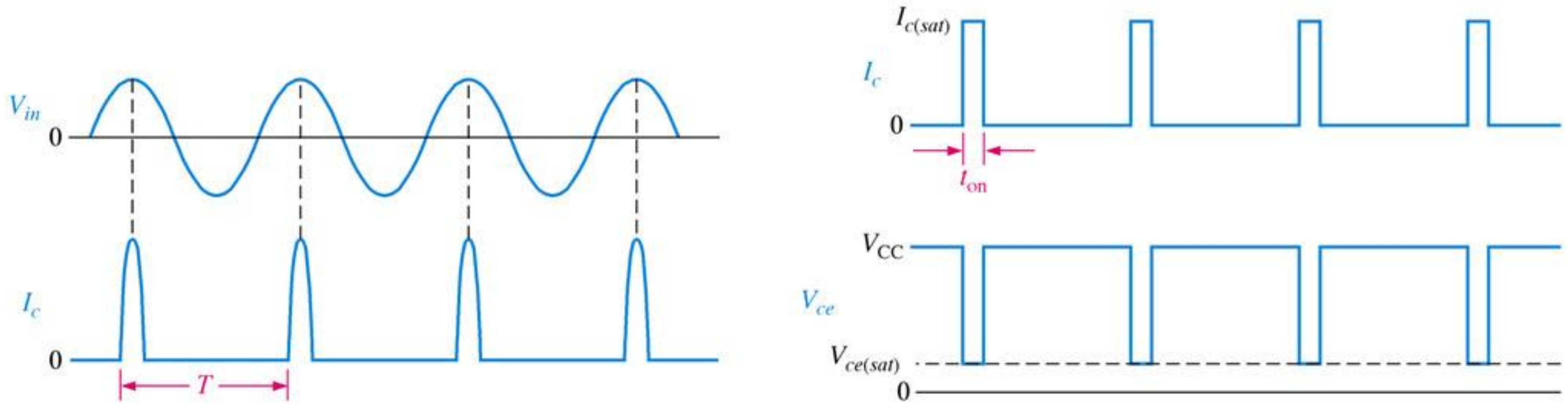


FIGURE 9-24 Class C waveforms.



(a) Collector current pulses

(b) Ideal class C waveforms

FIGURE 9-25 Tuned class C amplifier.

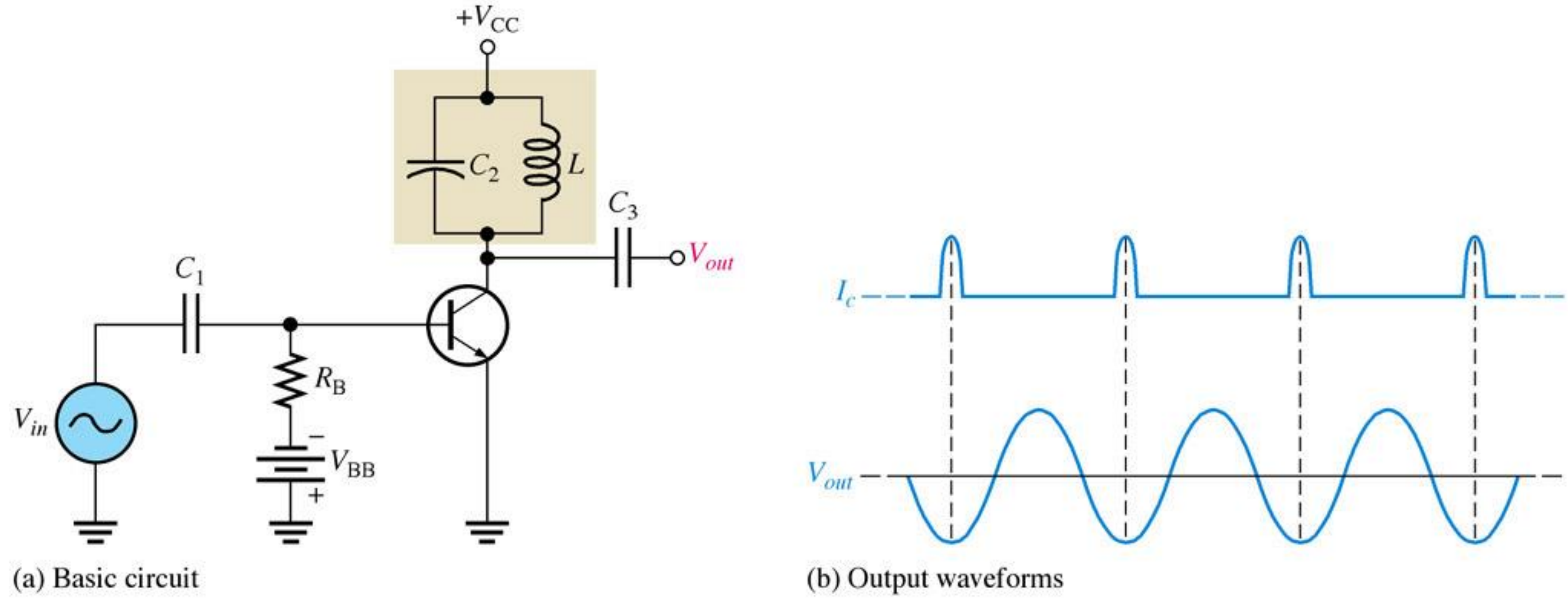
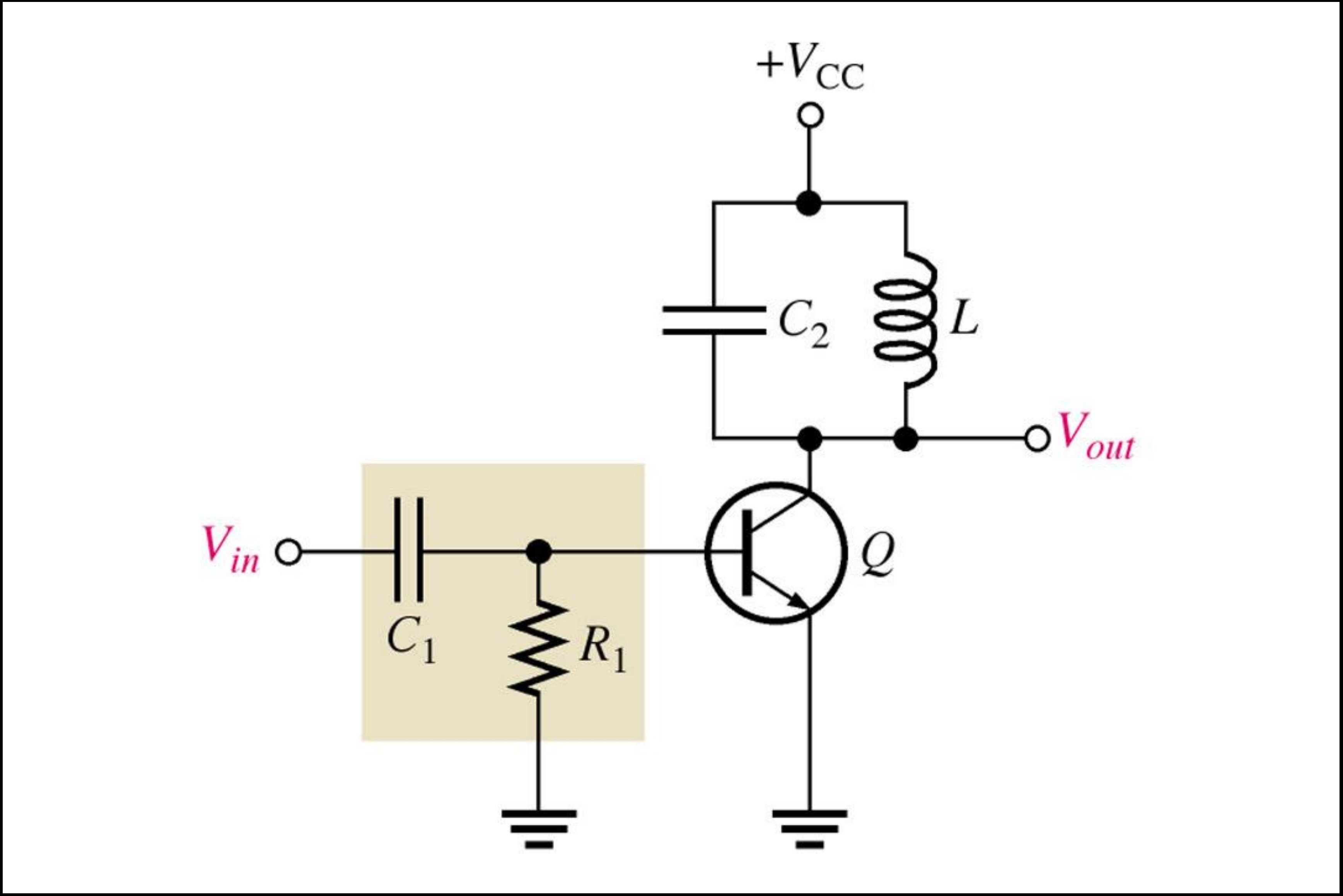
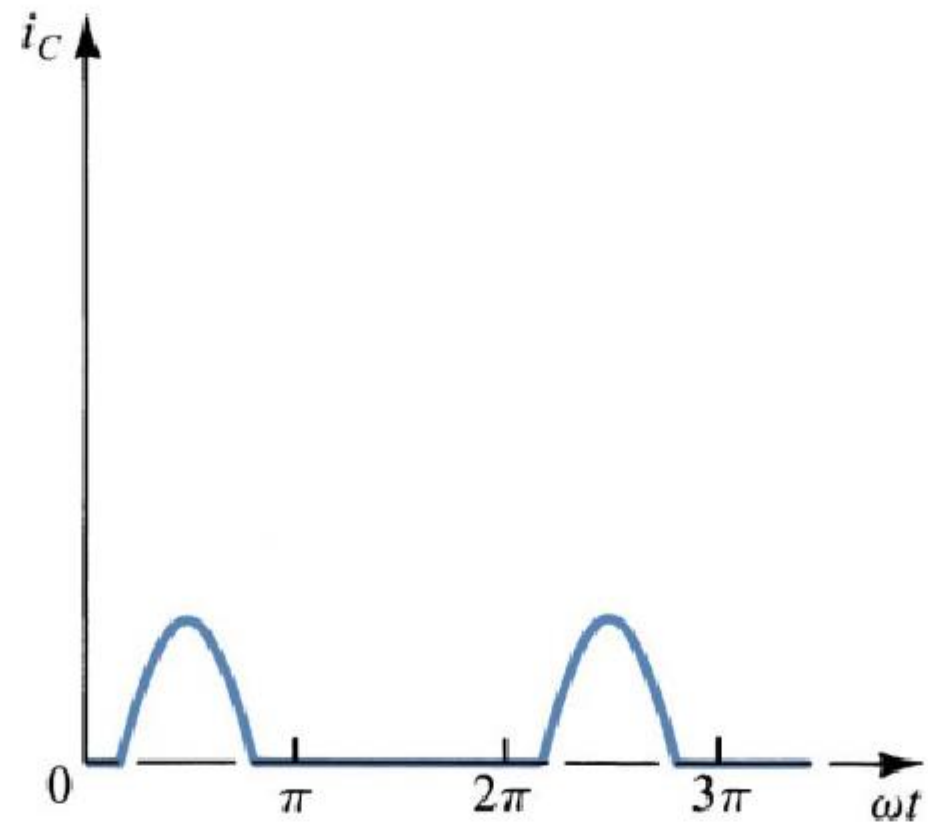


FIGURE 9-28 Tuned class C amplifier with clamper bias.





Collector current waveforms for transistors operating in class C amplifier stage

Class C

- However, the RF circuits where Class C amps are used, employ filtering so that the final signal is completely acceptable.

- Class C amps are **quite efficient**.